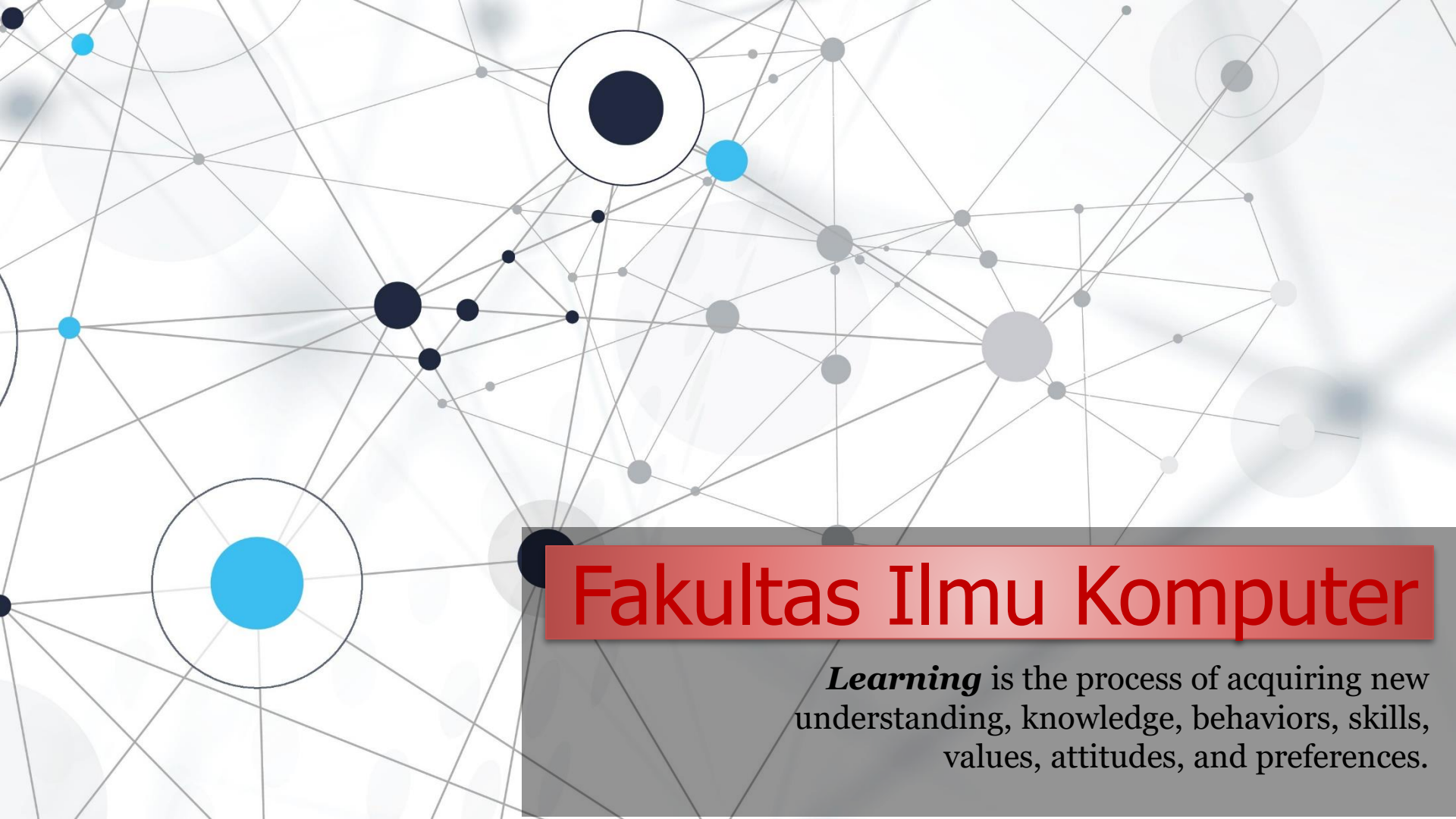


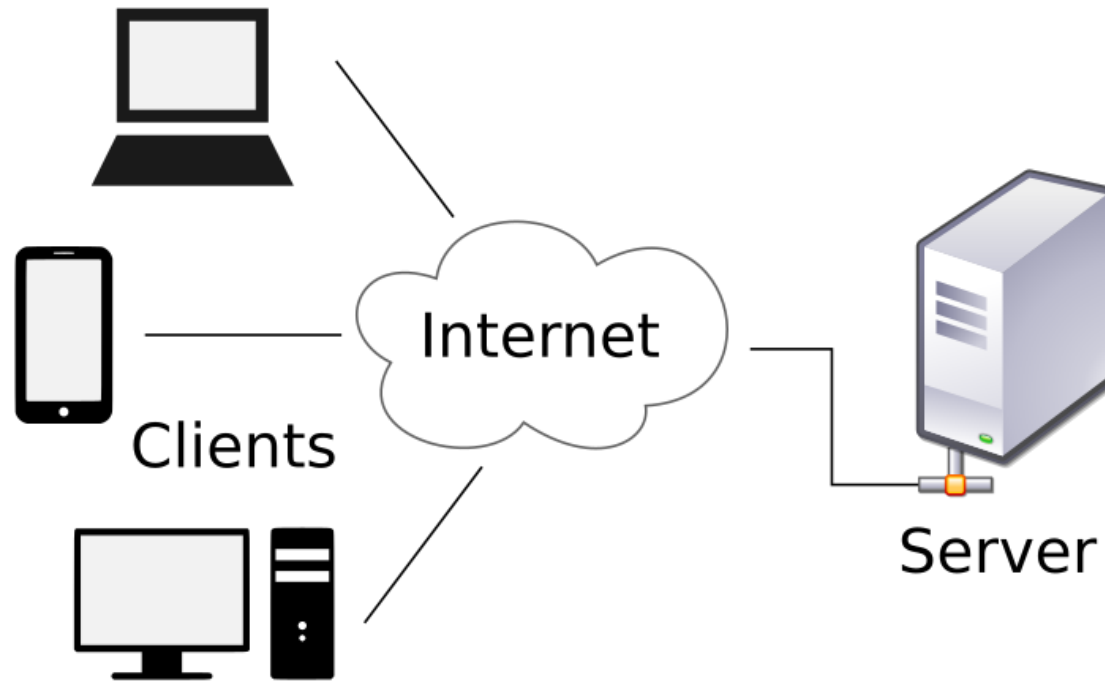
An abstract network diagram with various sized nodes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

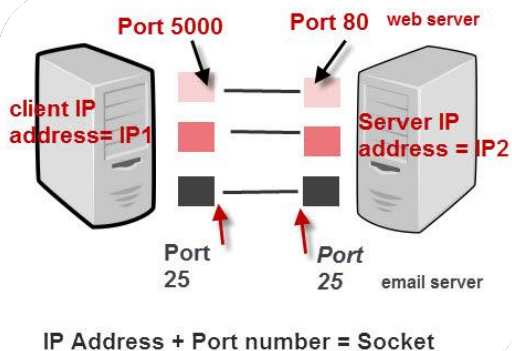
AJAX for PHP Web Applications

Client Server Communications

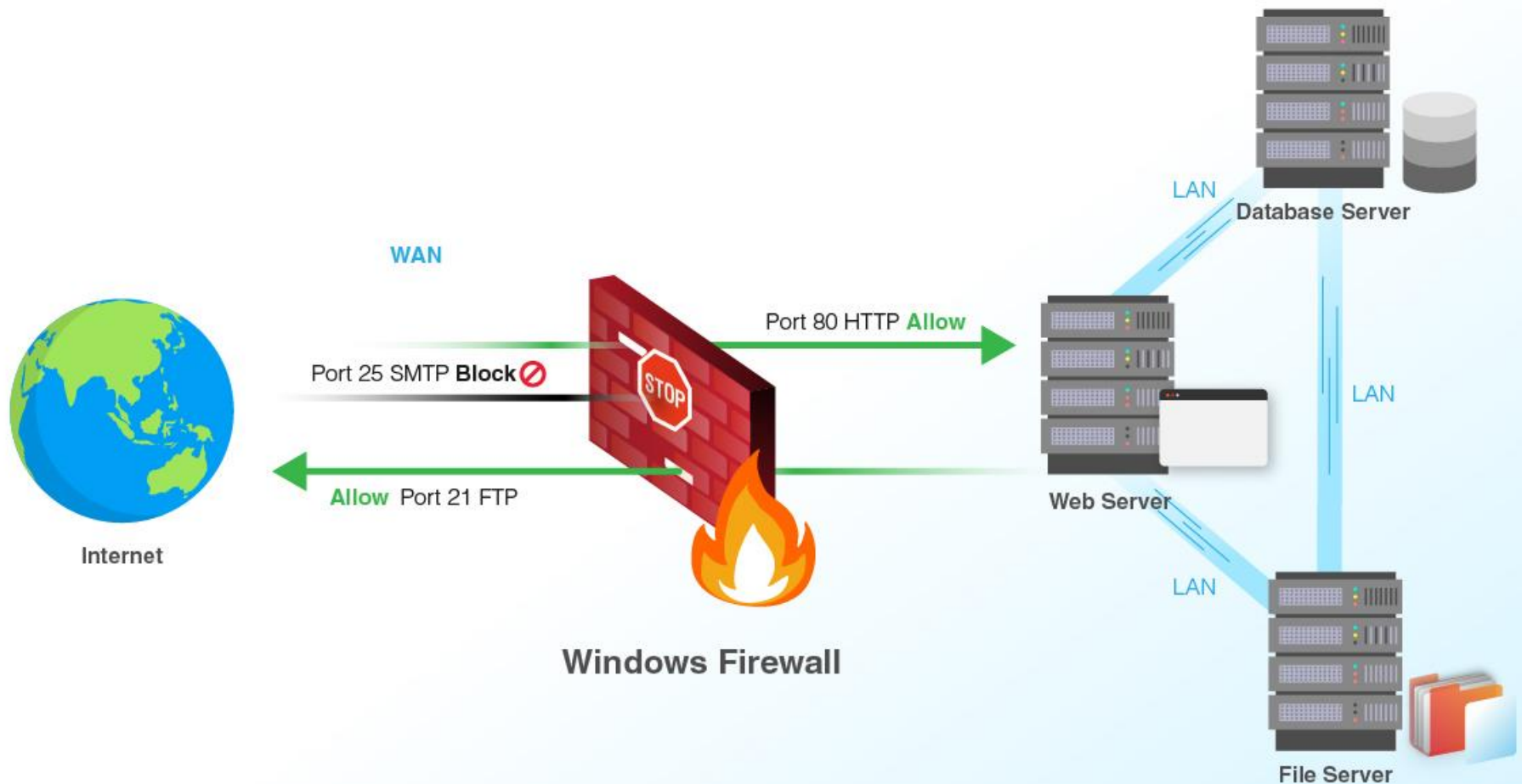


Servers are devices or software applications that provide services, data, or resources to clients. **Servers** are designed to (1) listen for incoming requests, (2) process those requests, and (3) send back appropriate responses/output.

[1] Server Ports



A server port is a **virtual doorway** or we called “*logical endpoint*” on a server that acts as for **specific applications or services**, identified by a unique number (0-65535) to ensure data is directed to the correct destination.



A web server port is used for communication endpoint, these port can be represented by a number from 0 to 65,535, that directs incoming internet traffic to the correct application or service on a server.

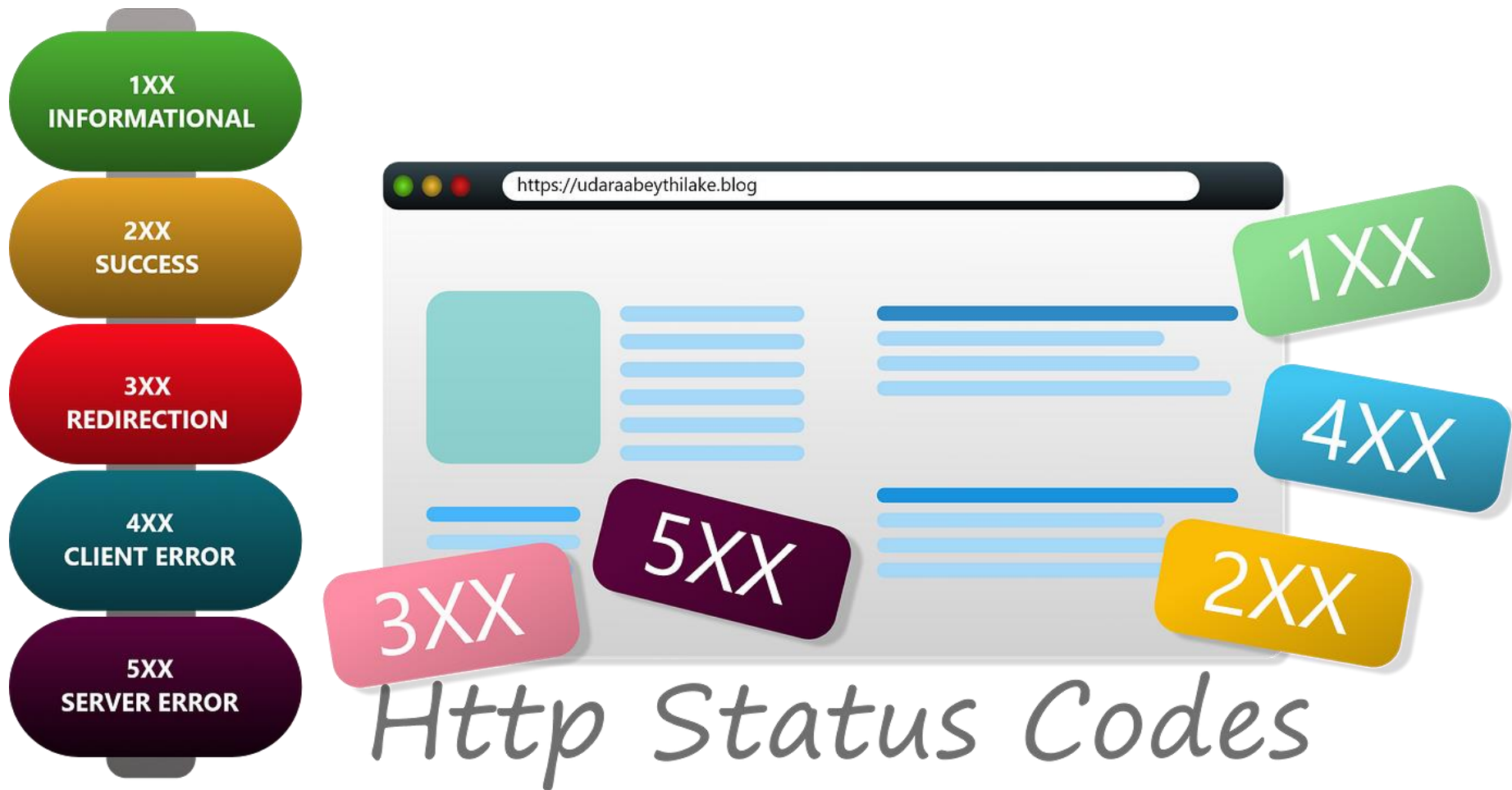
[2] HTTP/HTTPS Protocol



Port 80: Hypertext Transfer Protocol (**HTTP**)

Port 443: Hypertext Transfer Protocol Secure (**HTTPS**)

HTTP & HTTPS are used for web browsing and communication between web browsers and web servers. **HTTP is the standard protocol**, while **HTTPS adds security through encryption**.



[3] HTTP Status Codes are three-digit codes returned by the server in response to a request from a client (browser, API, or other application). These codes help in understanding whether the request was successful, encountered an error, or requires further action.

HTTP Status Codes

Successful



✓ GET, HEAD, PUT, POST

✓ 200 OK

201 Created (for POST)

204 No Content



Redirection



301 Moved Permanently

302 Found (temporarily)



Client Error



Incorrect Request

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found



Server Error



500 Internal Server Error

501 Not Implemented

502 Bad Gateway

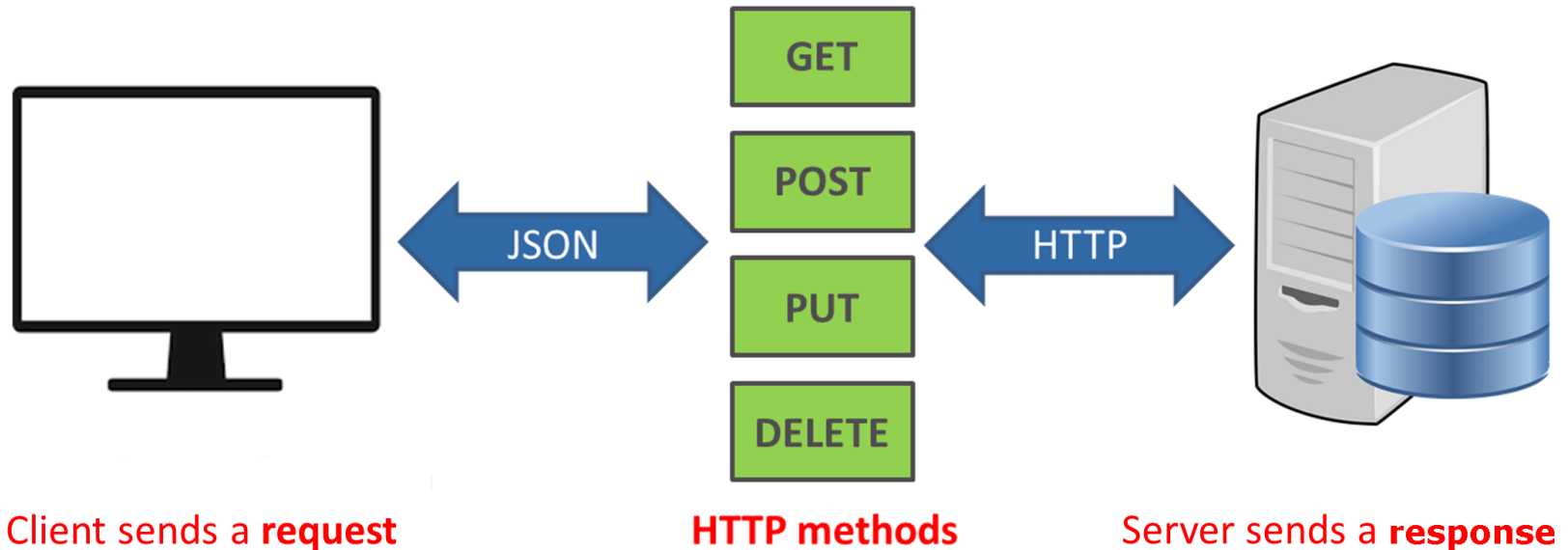
503 Service unavailable

504 Gateway Timeout



HTTP Status
Codes are three-
digit codes
returned by the
server

[4] Client Request Methods



Representational State Transfer (REST) is an architectural style for designing networked applications. It involves using standard HTTP methods (GET, POST, PUT, DELETE) to interact with resources on servers. RESTful APIs provide a structured way for clients to communicate with servers, typically exchanging data in JSON or XML format.



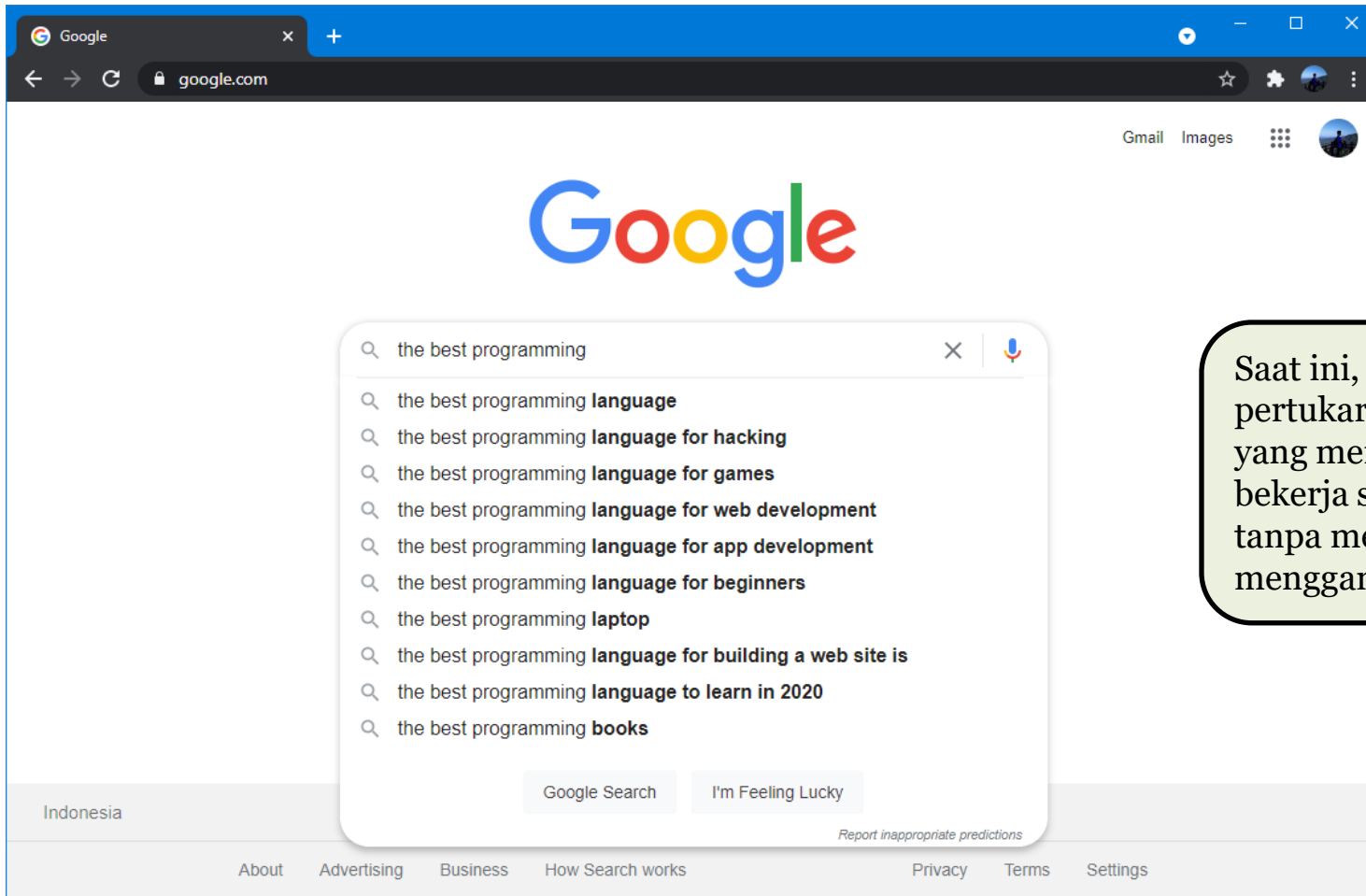
Ajax (programming)

Autocomplete Suggestion

Apa yang dimaksud dengan fitur *autocomplete suggestion* di *website*?



Contoh Penggunaan AJAX



Saat ini, dengan adanya AJAX, pertukaran data dan layar yang menampilkan data akan bekerja secara bersamaan tanpa menginterfensi atau mengganggu fungsi yang lain.



Contoh fitur **Google Autocomplete**, ketika mengetik kata kunci (*keyword*) di Google, maka fitur dari system ini akan menyelesaikan secara otomatis. Meskipun keyword akan berubah-ubah secara **real time**, tampilan halamannya tetap sama. Pada awal tahun 90-an, di mana *internet* belum secanggih dan semasif sekarang, fitur tersebut mengharuskan Google untuk memuat halaman setiap kali rekomendasi kata kunci baru muncul di layar.

Ajax (*programming*)

(*esinkrones*)

Asynchronous

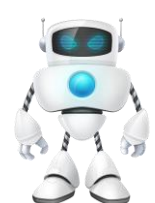
Ajax (short for “**A**synchronous **J**avaScript and **X**ML”) is a set of web development techniques using many web technologies on the **client-side** to create *asynchronous web applications*.

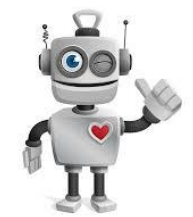
With Ajax, **web applications** can send and retrieve data from a server asynchronously (*in the background*) without interfering with the display and behaviour of the existing *page*.

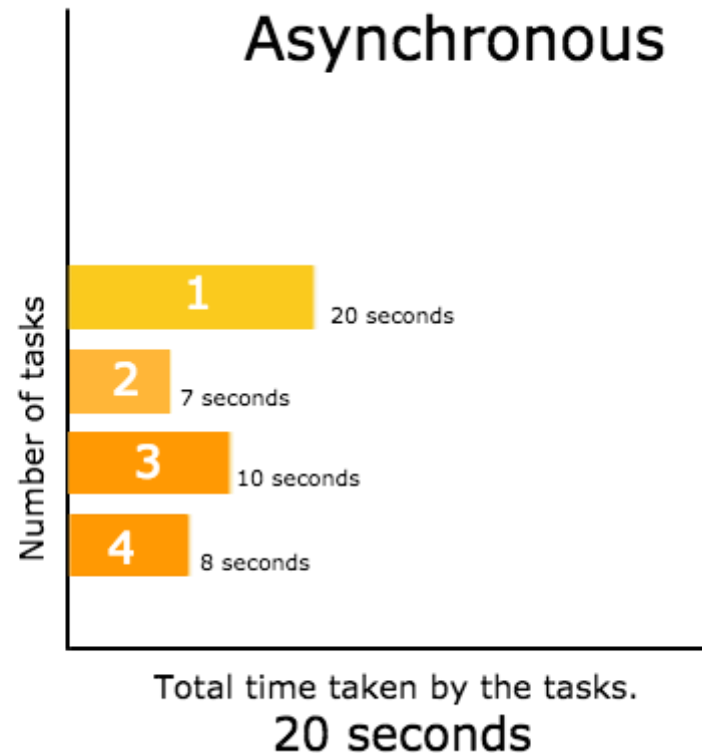
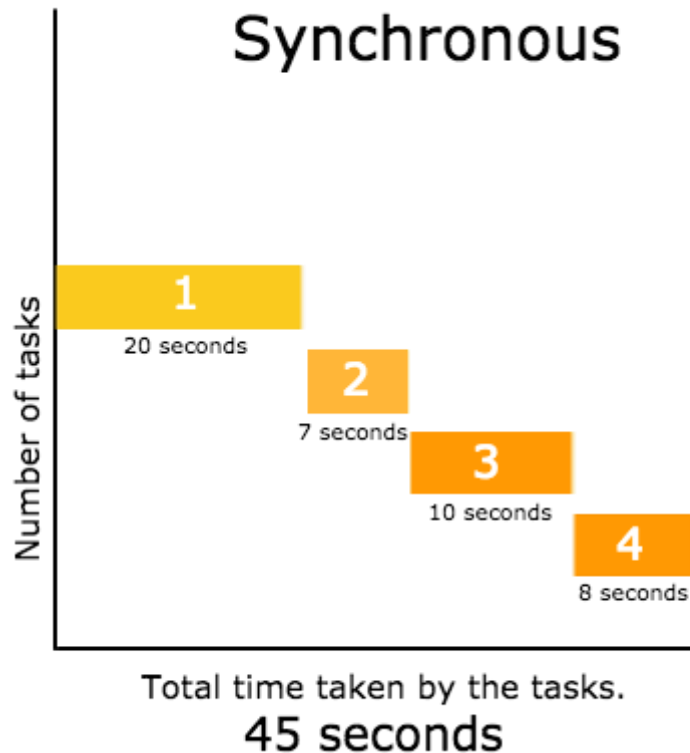
Asynchronous Javascript and XML (Ajax)

- 
- ❑ Mengacu pada sekumpulan teknis pengembangan web (*web development*) yang memungkinkan aplikasi web untuk bekerja secara **asynchronous (tidak langsung)**, memproses setiap *request (permintaan)* yang datang ke *server* di sisi **background**.

- 
- ❑ **JavaScript** merupakan bahasa coding yang kerap digunakan (*salah satu fungsinya*) untuk ***mengelola konten dinamis website*** dan memungkinkan ***interaksi user yang dinamis***.

- 
- ❑ HTML, XML (*eXtensible Markup Language*) adalah varian lain dari bahasa *markup*. Jika **HTML dirancang untuk menampilkan data**, maka **XML dirancang untuk memuat dan membawa data**.

- 
- ❑ **JavaScript dan XML bekerja secara asynchronous di dalam AJAX**. Maka, aplikasi web yang menggunakan AJAX dapat mengirimkan dan menerima data dari *server* ***tanpa harus me-reload keseluruhan halaman website***.



Perbedaan Web Javascript Synchronous *dengan* Asynchronous:

- ❖ **Synchronous** setiap proses di panggil maka akan menunggu dahulu proses tersebut baru proses selanjutnya dapat di *eksekusi*.
- ❖ **Asynchronous** merupakan kebalikan dari proses Synchronous dimana proses **Asynchronous** akan menjalankan proses tanpa menunggu proses yang sebelumnya di panggil selsesai di proses.

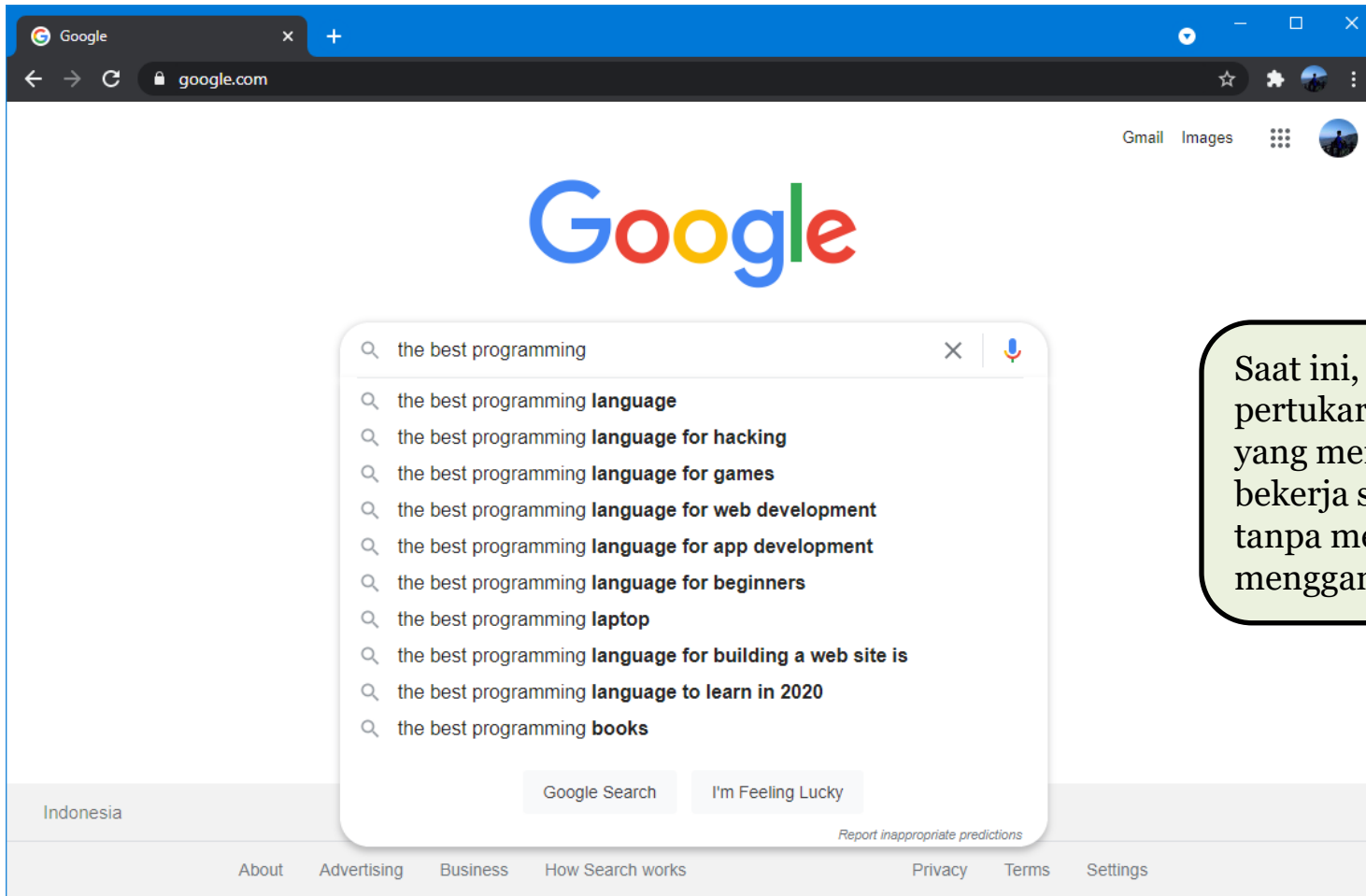
Asynchronous Javascript and XML (*Ajax*)

Dengan AJAX memungkinkan *kita* untuk dapat menerima dan mengirim data/informasi dalam format JSON, XML, HTML dan TEXT tanpa perlu melakukan ***refresh halaman web***.

- ❖ The \$.ajax() method in jQuery is used to perform **asynchronous requests** to a web server or other external data source.
- ❖ \$.ajax() method is one way to send HTTP GET, POST, PUT, DELETE, etc. requests, and receive a response from the server without the need to replace the entire web page.
- ❖ \$.ajax() method is often used to **interact with web APIs**, **retrieve data from the server**, **send data to the server**, and **perform other operations** that involve **communicating with the server** without the need to refresh the page.



Contoh Penggunaan AJAX



Saat ini, dengan adanya AJAX, pertukaran data dan layar yang menampilkan data akan bekerja secara bersamaan tanpa menginterfensi atau mengganggu fungsi yang lain.



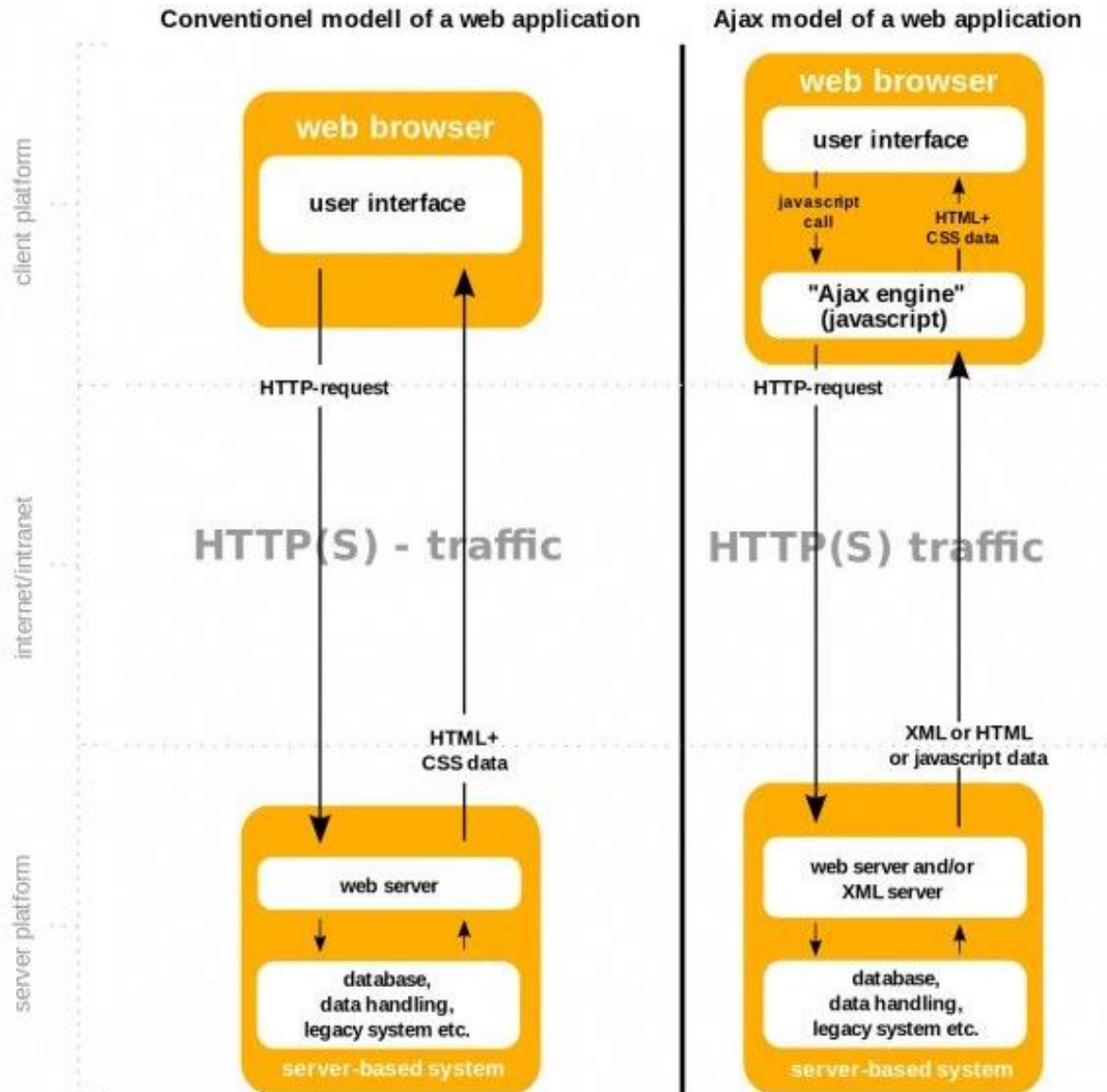
Contoh fitur **Google Autocomplete**, ketika mengetik kata kunci (*keyword*) di Google, maka fitur dari system ini akan menyelesaikan secara otomatis. Meskipun keyword akan berubah-ubah secara **real time**, tampilan halamannya tetap sama. Pada awal tahun 90-an, di mana *internet* belum secanggih dan semasif sekarang, fitur tersebut mengharuskan Google untuk memuat halaman setiap kali rekomendasi kata kunci baru muncul di layar.

Cara Kerja AJAX

AJAX adalah sekumpulan teknik *pengembangan web*, terdiri atas:

- ✓  **HTML/XHTML** sebagai bahasa utama dan **CSS (*Cascading Style Sheets*)** untuk menampilkan data.
- ✓  The **Document Object Model (DOM)** untuk menampilkan data yang dinamis beserta interaksinya.
- ✓  **XML** untuk pertukaran data, sedangkan **XSLT (*eXtensible Stylesheet Language Transformations*)** untuk manipulasi data. Sebagian besar developer mulai mengganti XML dengan **JSON (*JavaScript Object Notation*)** karena bentuknya yang mendekati JavaScript.
- ✓  Objek **XMLHttpRequest** untuk komunikasi tidak langsung (*asynchronous*).
- ✓  Bahasa pemrograman **JavaScript** untuk menyatukan semua teknologi.

Diagram Ajax



Di background,
web browser
mengirimkan
permintaan
(*request*) HTTP ke
server.

Perbandingan Konsep Konvensional & AJAX

Konsep Konvensional (biasa/lazim):

- 1) Permintaan (*request*) HTTP dikirimkan dari *web browser* ke *server*.
- 2) Server menerima, dan kemudian, mengambil data.
- 3) Server mengirimkan data yang diminta ke *web browser*.
- 4) *Web browser* menerima data dan mereload halaman untuk menampilkan data.

Konsep AJAX:

- 1) Browser membuat **JavaScript call** yang kemudian akan mengaktifkan XMLHttpRequest.
- 2) Di **background**, *web browser* mengirimkan permintaan (*request*) HTTP ke server.
- 3) Server menerima, mengambil, dan mengirimkan data kembali ke *web browser*.
- 4) Web browser menerima data yang diminta dan akan langsung ditampilkan di halaman **tanpa harus melalui proses reload terlebih dulu**.

Fungsi AJAX adalah untuk mempersingkat atau mempermudah *user experience*.

Questions



Bagaimana cara untuk menyembunyikan IP address computer *klien* ketika berkunjung ke halaman website yang dikunjungi ?

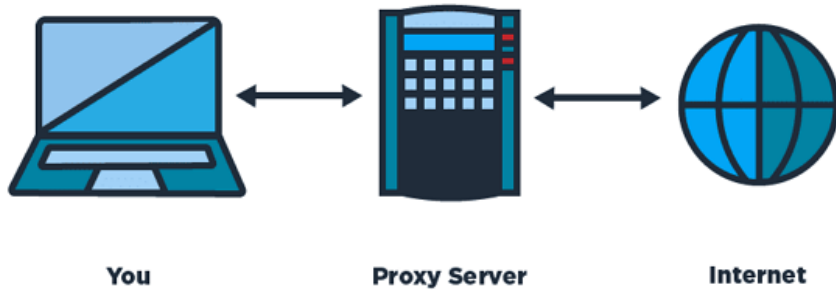
How to hide the client computer's IP address when visiting a web page?



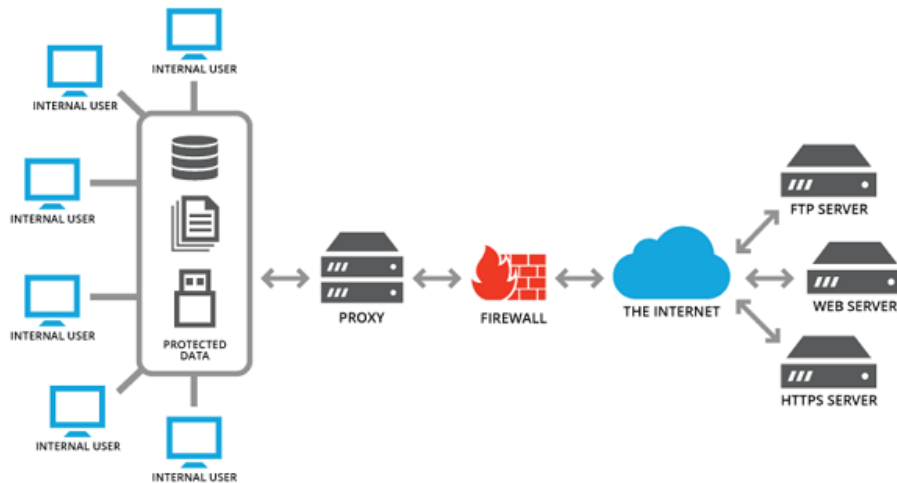
Bagaimana cara sehingga kita bisa *streaming* sebuah *video* dengan cepat? Padahal video terlalu besar? Prosesnya kira-kira bagaimana?

How to stream a video quickly? Even though the video size is too big?

Proxy Server



Proxy adalah sebuah *server komputer* yang berfungsi meneruskan permintaan pengguna ke server lain di internet. Ibaratnya, *proxy server* merupakan penghubung antara komputer pribadi dengan jaringan internet.



Beberapa macam kegunaan proxy server:

- ❑ **Untuk menyembunyikan alamat IP komputer klien/pengguna**, sehingga siapapun bisa berselancar secara anonim tanpa meninggalkan jejak alamat IP di halaman website yang dikunjungi. Misalnya, jika seorang klien mengunjungi Google melalui server proxy, maka Google akan melihat bahwa yang mengunjunginya merupakan *proxy server*, bukan klien, sehingga Google pun tidak dapat melacak posisi klien pengguna tersebut.
- ❑ **Berselancar lebih cepat di internet**. Semua permintaan dari komputer klien akan mencapai proxy server terlebih dahulu. Kemudian proxy akan mengecek apakah halaman web yang dikunjungi pengguna tersebut sudah memiliki caching yang tersimpan atau belum. Jika ada, maka client akan mendapatkan *feedback* dari data caching tersebut, sehingga klien dapat berselancar lebih cepat di internet.
- ❑ **Untuk memblokir situs yang tak diinginkan**. Misalkan untuk jaringan internet di perkantoran, agar pekerja di kantor tersebut tidak memanfaatkan internet untuk bermain media sosial dan menonton, maka beberapa situs bisa dimasukkan dalam blacklist proxy sehingga tidak dapat diakses melalui jaringan kantor tersebut.



Melihat proses AJAX

Kita bisa melihat proses AJAX melalui *inspect elemen di web browser*, lalu buka tab Network dan aktifkan filter XHR (XML HTTP Request).

Developer Tools - Petani Kode: Programmer Pengguna Linux - <https://www.petanikode.com/>

Inspector Console Debugger Style Editor Performance Memory **Network** Storage >> << <<< >>>

Filter URLs || ☐ Persist Logs ☐ Disable cache No throttling HAR

All HTML CSS JS **XHR** Fonts Images Media WS Other

Status	Method	Domain	File	Cause	Type	Transferred	Size	0 ms	5.12 s	10.24 s	15.36 s
200	GET	l.sharethis...	pview?...	xhr	plain	476 B	51 B	8073 ms			
200	POST	nondescriptcr...	ulmDoi...	fetch	json	780 B	208 B	1426 ms			
200	GET	web.facebo...	/ajax/b...	xhr	js	12.18 KB	28.58 KB	311 ms			
200	POST	web.facebo...	bz	xhr	js	804 B	130 B	656 ms			

Metode pengiriman Data ke server.
GET artinya ambil dan POST artinya kirim

Ini alamat tujuan pengiriman dan pengambilan data

4 requests 28.96 KB / 14.19 KB transferred Finish: 12.61 s load: 7.06 s

Note: AJAX juga mendukung format JSON, Plain Text dan HTML.



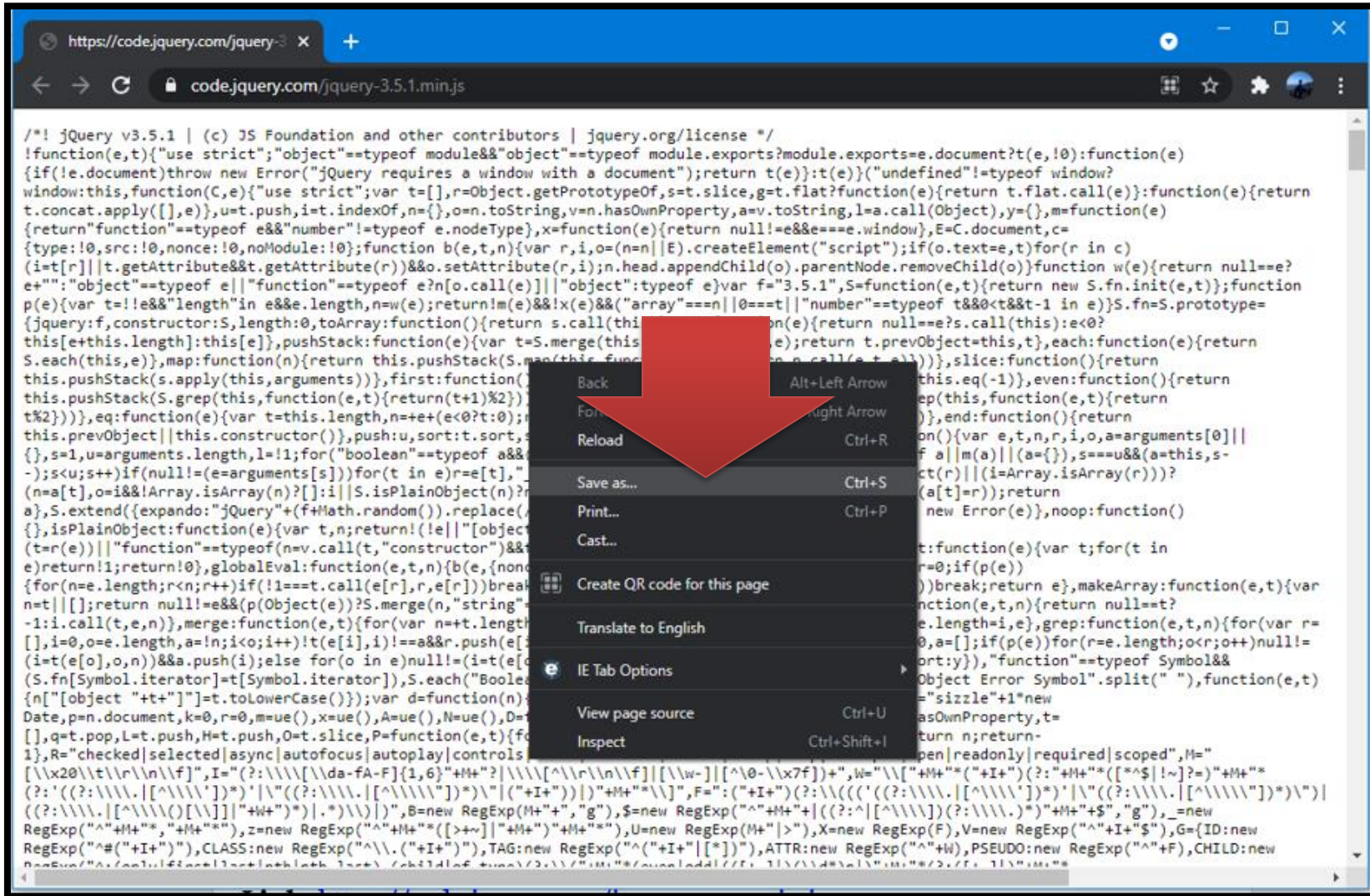
AJAX dengan jQuery & PHP



jQuery is a JavaScript library designed to simplify HTML DOM (*Document Object Model*) tree traversal and manipulation, as well as **event handling**, **CSS animation**, and **Ajax**. It is free, open-source software using the permissive MIT License. As of May 2019, jQuery is used by 73% of the 10 million most popular websites. - *Wikipedia*

Download jQuery-3.5.1

Belajar ajax dasar menggunakan jQuery dan PHP



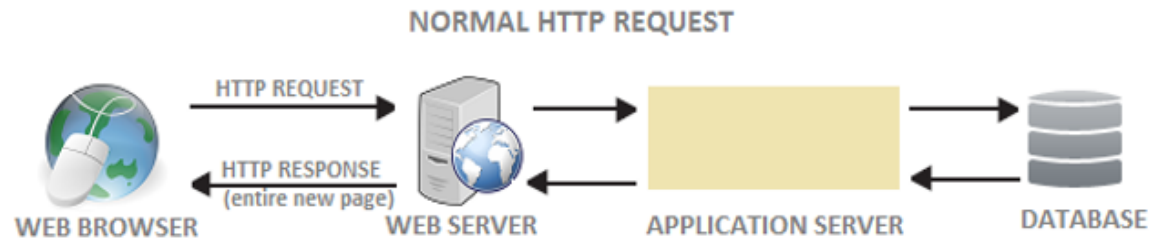
Link: <https://code.jquery.com/jquery-3.5.1.min.js>

jQuery AJAX Methods

AJAX is the art of exchanging data with a server, and update parts of a web page - without reloading the whole page.

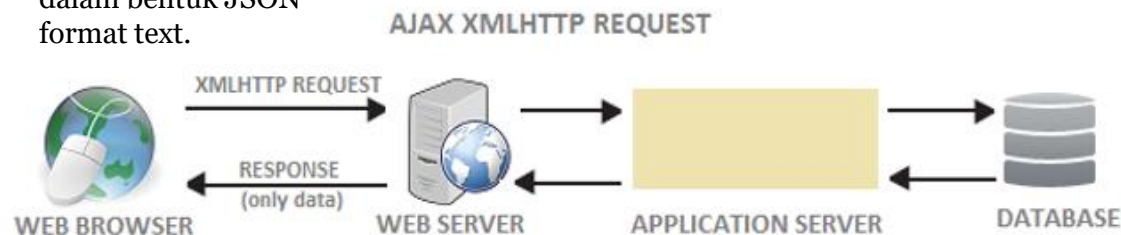
Method	Description
<u>\$.ajax()</u>	Performs an async AJAX request
\$.ajaxPrefilter()	Handle custom Ajax options or modify existing options before each request is sent and before they are processed by \$.ajax()
\$.ajaxSetup()	Sets the default values for future AJAX requests
\$.ajaxTransport()	Creates an object that handles the actual transmission of Ajax data
<u>\$.get()</u>	Loads data from a server using an AJAX HTTP GET request
\$.getJSON()	Loads JSON-encoded data from a server using a HTTP GET request
\$.parseJSON()	Deprecated in version 3.0, use JSON.parse() instead. Takes a well-formed JSON string and returns the resulting JavaScript value
\$.getScript()	Loads (and executes) a JavaScript from a server using an AJAX HTTP GET request
\$.param()	Creates a serialized representation of an array or object (can be used as URL query string for AJAX requests)
<u>\$.post()</u>	Loads data from a server using an AJAX HTTP POST request
ajaxComplete()	Specifies a function to run when the AJAX request completes
ajaxError()	Specifies a function to run when the AJAX request completes with an error
ajaxSend()	Specifies a function to run before the AJAX request is sent
ajaxStart()	Specifies a function to run when the first AJAX request begins
ajaxStop()	Specifies a function to run when all AJAX requests have completed
ajaxSuccess()	Specifies a function to run when an AJAX request completes successfully
<u>load()</u>	Loads data from a server and puts the returned data into the selected element
serialize()	Encodes a set of form elements as a string for submission
serializeArray()	Encodes a set of form elements as an array of names and values

What about jQuery & AJAX?



HTTP requests can be made using various technologies in the web environment, such as **HTML forms (for the POST method)**, JavaScript **fetch API**, or by using other technologies that follow the HTTP standard.

Data yang dikirim
dalam bentuk JSON
format text.



AJAX (Asynchronous JavaScript and XML) XMLHttpRequest is a special technology used in JavaScript to **make asynchronous HTTP requests** from a web browser to a server **without the need to refresh the page**. It is one of the most commonly used ways to make asynchronous HTTP requests in web development.

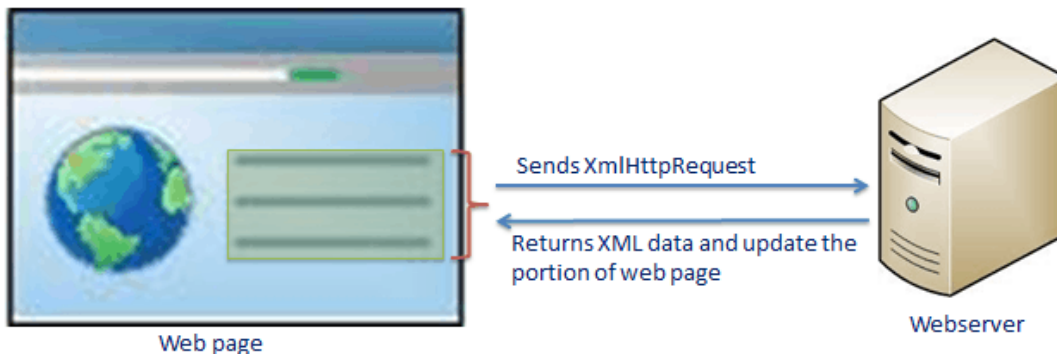
What about jQuery & AJAX?



- 1) jQuery provides several methods for AJAX functionality. With the jQuery AJAX methods, we can request text, HTML, XML, or JSON from a remote server using both HTTP Get and HTTP Post, and we can load the external data directly into the selected HTML elements of your web page.
- 2) Without jQuery, AJAX coding can be a bit tricky. Writing regular AJAX code can be a bit tricky, because different browsers have different syntax for AJAX implementation. This means that we will have to write extra code to test for different browsers.
- 3) The jQuery team has taken care of this, so we can write AJAX functionality with only one single line of code.

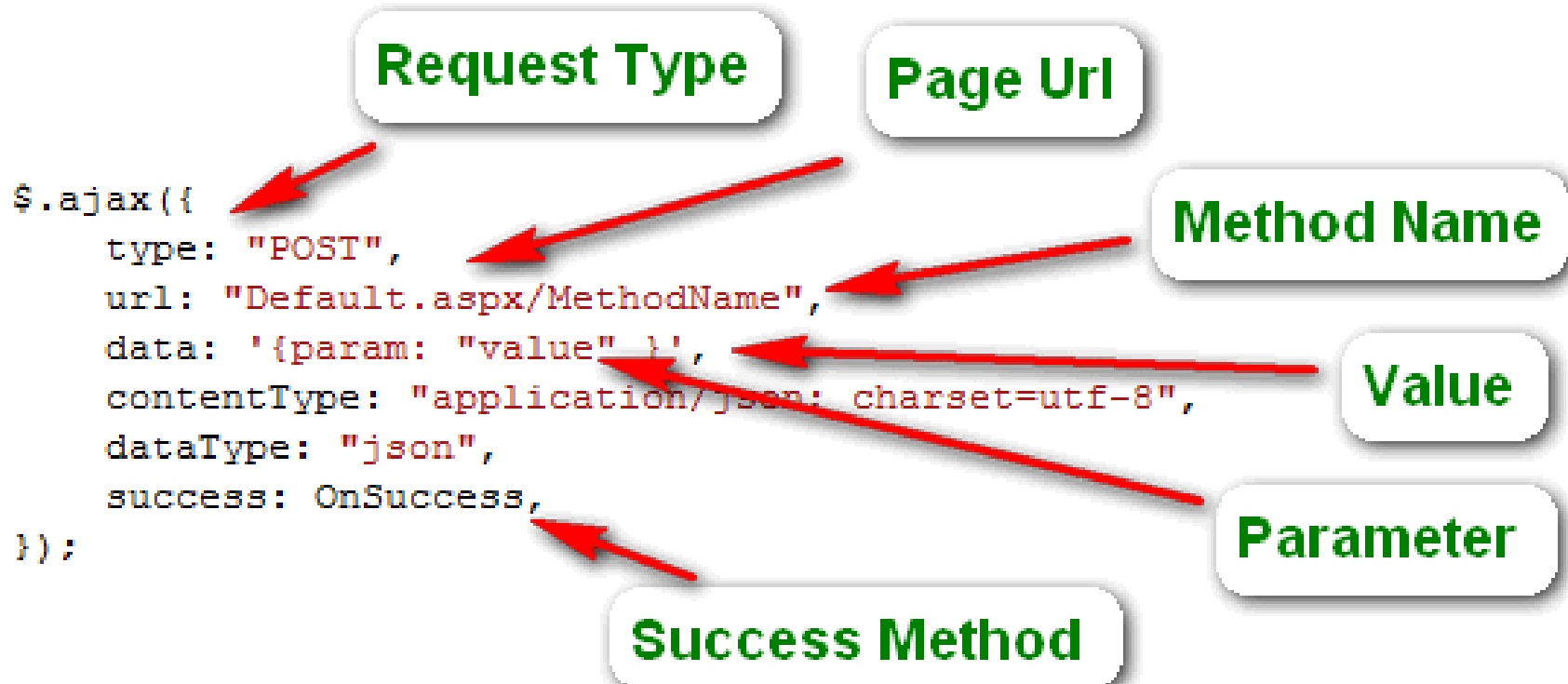
jQuery AJAX Methods

jQuery Methods	Ajax	Description
ajax()		Sends asynchronous http request to the server.
get()		Sends http GET request to load the data from the server.
Post()		Sends http POST request to submit or load the data to the server.
getJSON()		Sends http GET request to load JSON encoded data from the server.
getScript()		Sends http GET request to load the JavaScript file from the server and then executes it.
load()		Sends http request to load the html or text content from the server and add them to DOM element(s).

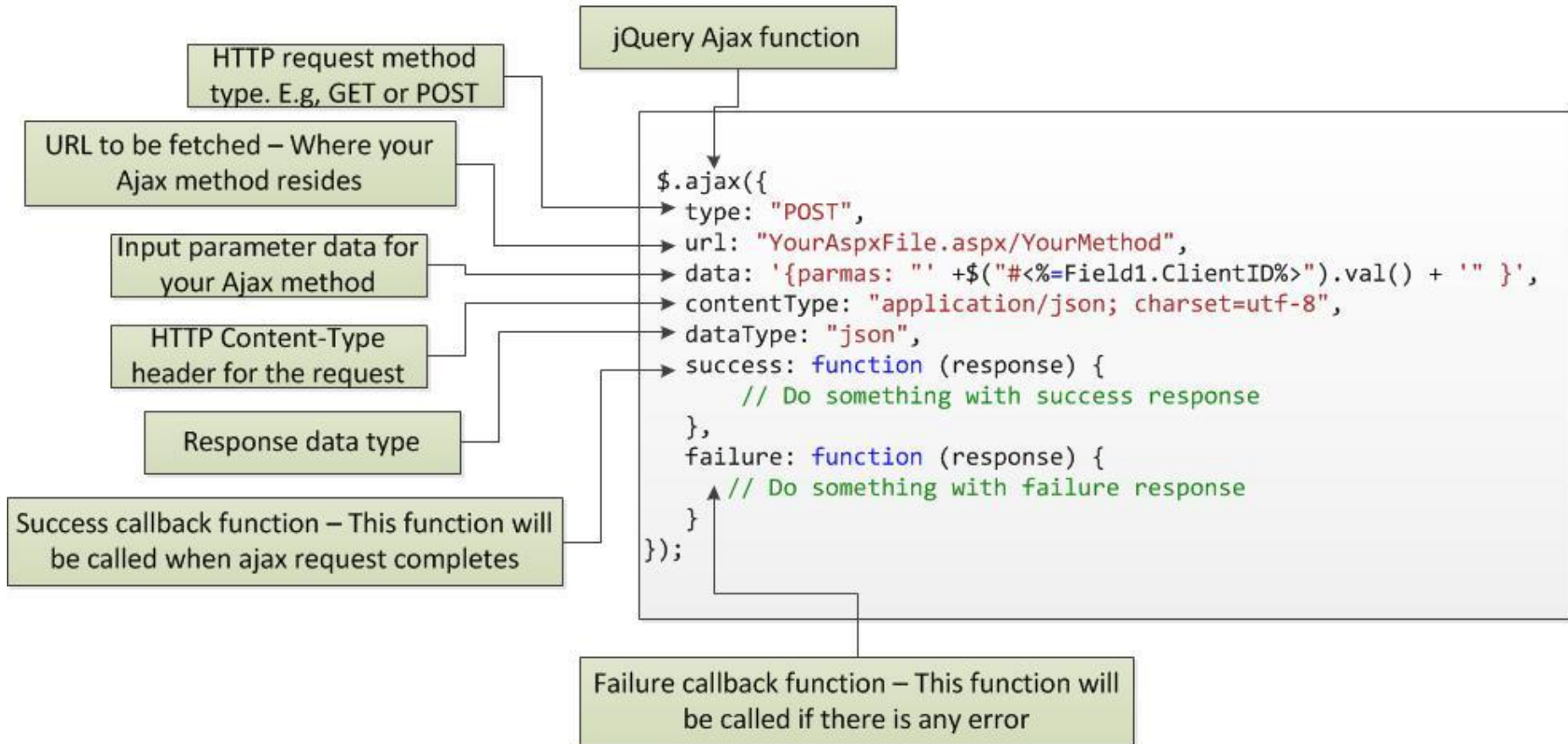


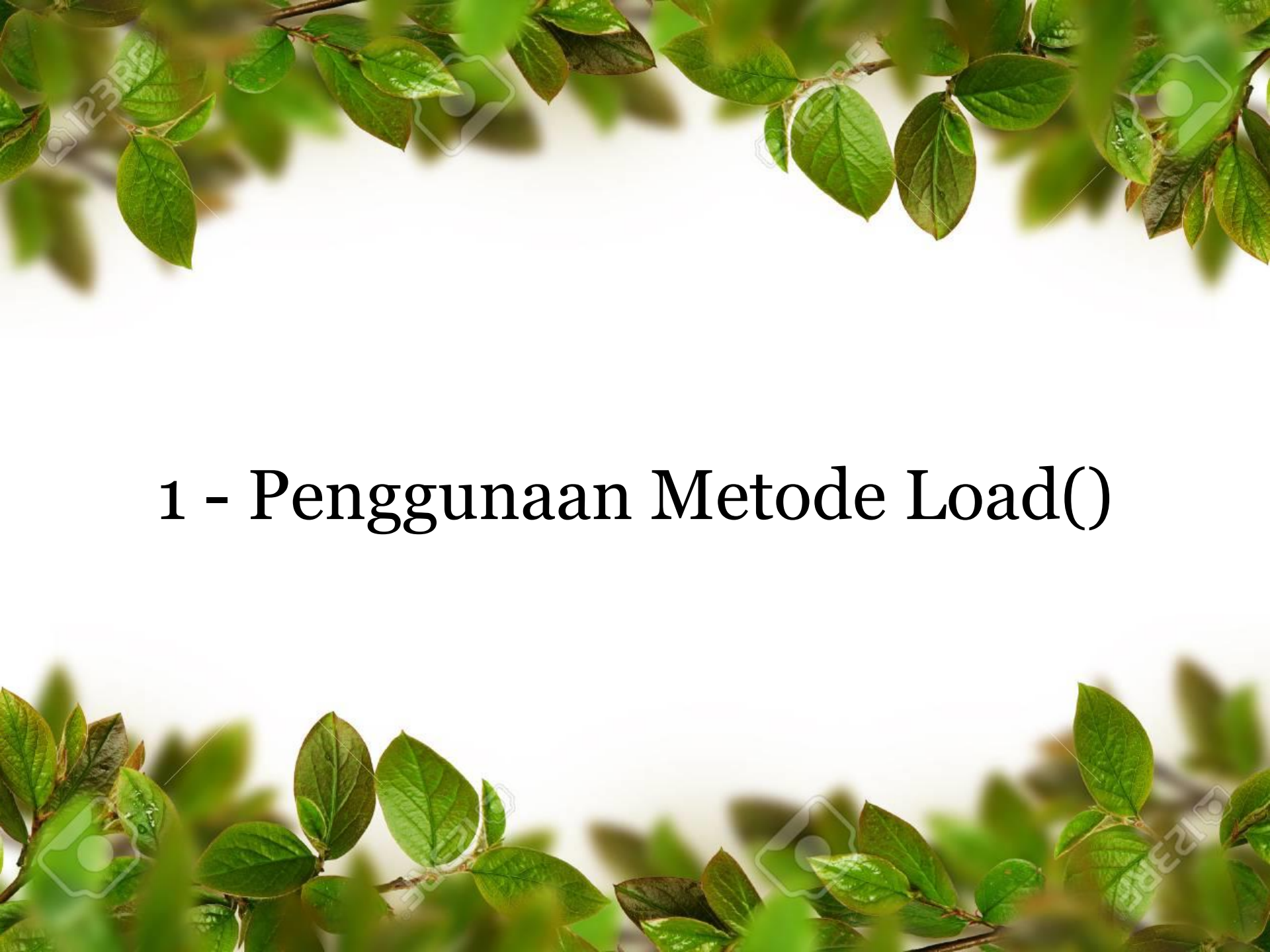
The jQuery library includes various methods to send Ajax requests. These methods internally use XMLHttpRequest object of JavaScript.

Example jQuery AJAX Methods



Example jQuery AJAX Methods





1 - Penggunaan Metode Load()

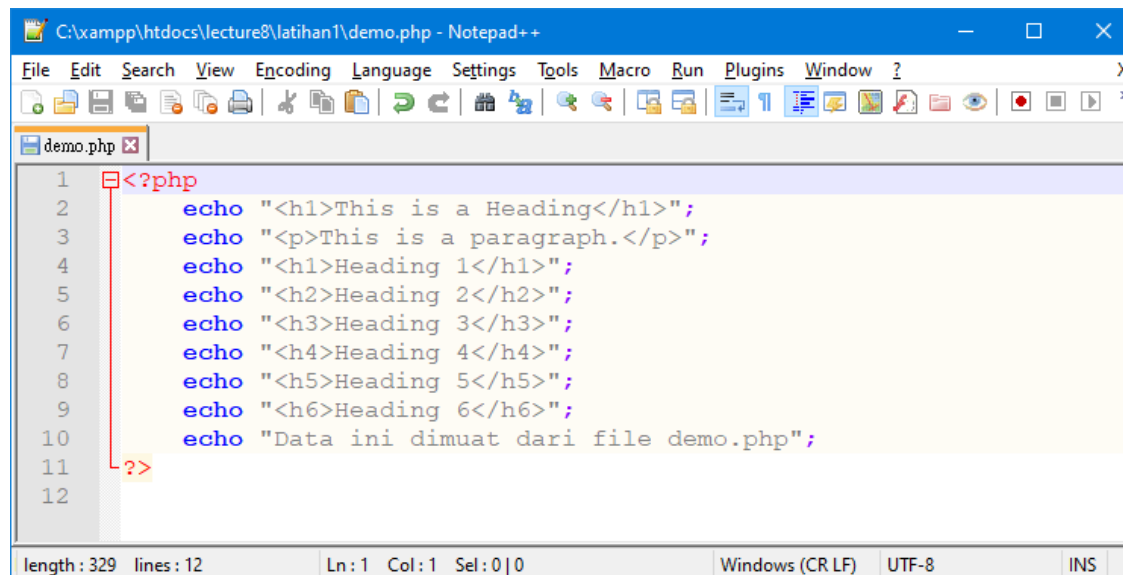
1 - Penggunaan Metode Load()

Metode **load()** adalah metode yang tersedia di dalam library jQuery yang digunakan untuk memuat data dari **server** yang kemudian dapat ditempatkan pada **elemen** tertentu di halaman **website**.

Bentuk Umum:

```
$(selector).load(URL,data,callback);
```

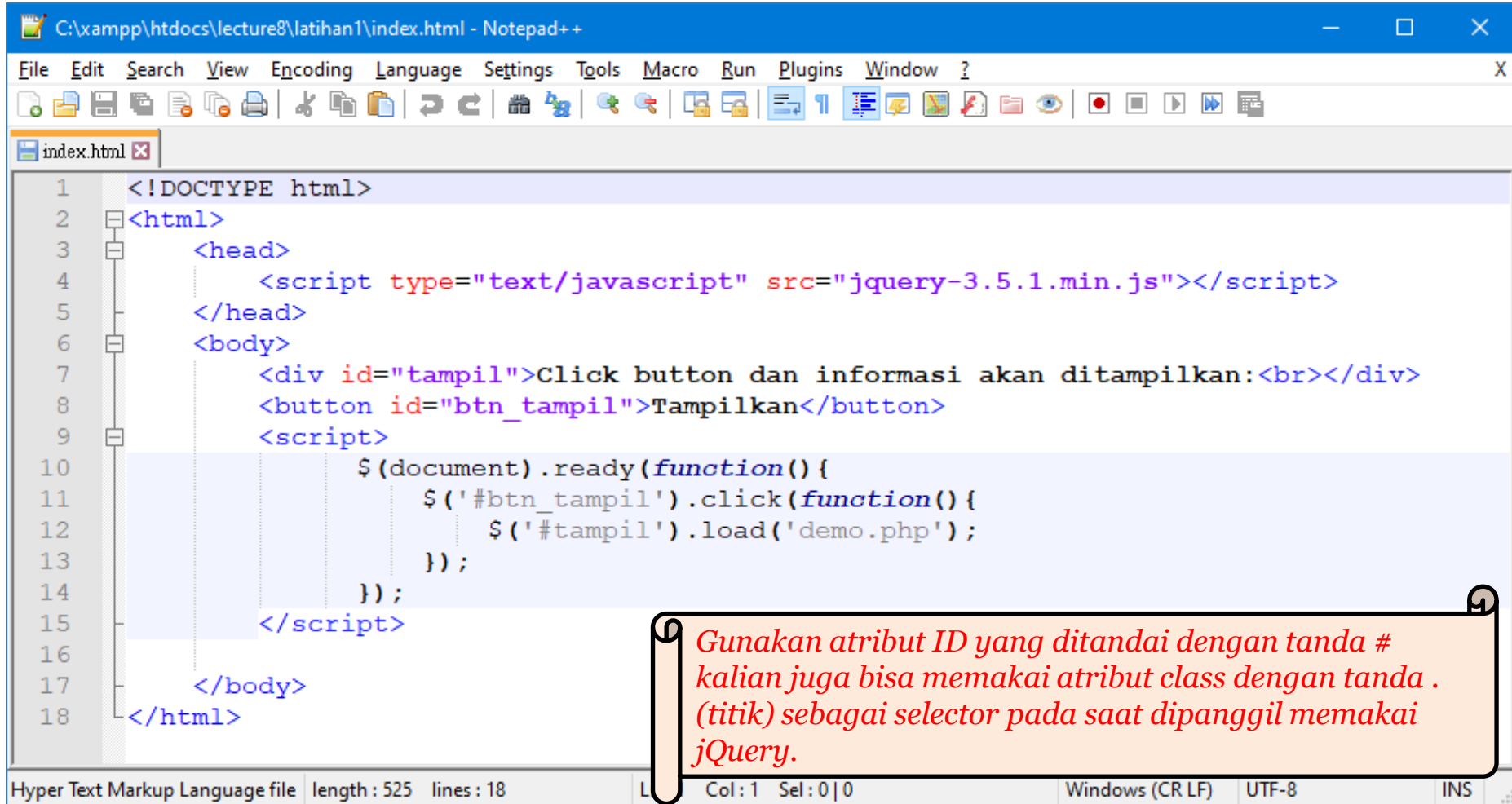
Langkah 1 : Buat file **demo.php**



```
C:\xampp\htdocs\lecture8\latihan1\demo.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
demo.php
1 <?php
2     echo "<h1>This is a Heading</h1>";
3     echo "<p>This is a paragraph.</p>";
4     echo "<h1>Heading 1</h1>";
5     echo "<h2>Heading 2</h2>";
6     echo "<h3>Heading 3</h3>";
7     echo "<h4>Heading 4</h4>";
8     echo "<h5>Heading 5</h5>";
9     echo "<h6>Heading 6</h6>";
10    echo "Data ini dimuat dari file demo.php";
11 ?>
12
length: 329 lines: 12 Ln: 1 Col: 1 Sel: 0|0 Windows (CR LF) UTF-8 INS
```

1 - Penggunaan Metode Load()

Langkah 2 : Buat file **index.php**



```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <script type="text/javascript" src="jquery-3.5.1.min.js"></script>
5   </head>
6   <body>
7     <div id="tampil">Click button dan informasi akan ditampilkan:<br></div>
8     <button id="btn_tampil">Tampilkan</button>
9     <script>
10      $(document).ready(function() {
11        $('#btn_tampil').click(function() {
12          $('#tampil').load('demo.php');
13        });
14      });
15    </script>
16  </body>
17 </html>
```

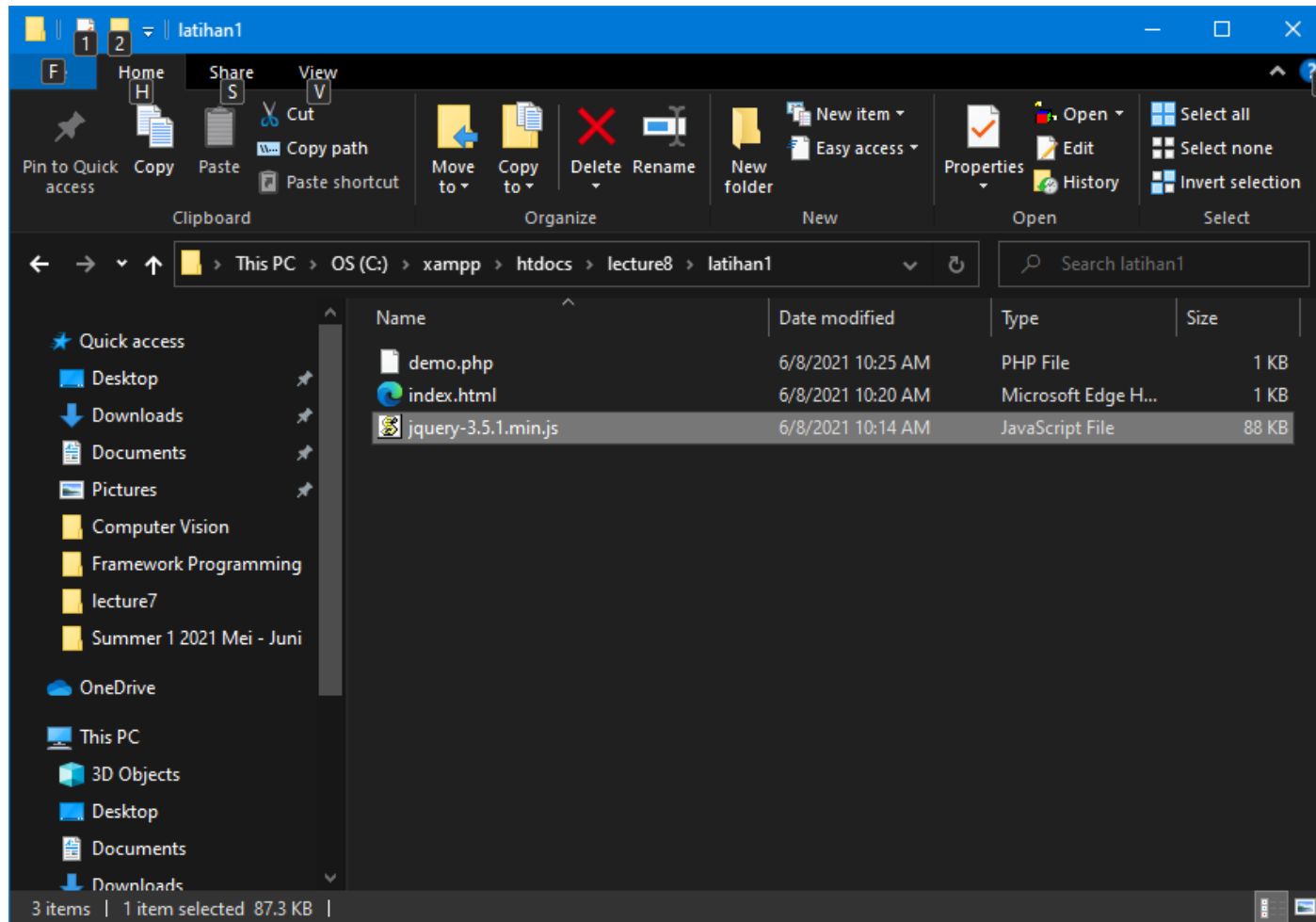
Gunakan atribut ID yang ditandai dengan tanda # kalian juga bisa memakai atribut class dengan tanda . (titik) sebagai selector pada saat dipanggil memakai jQuery.

File **index.php** berfungsi sebagai file utama yang akan memuat informasi yang di *request* menggunakan metode **load()** pada file **demo.php**.

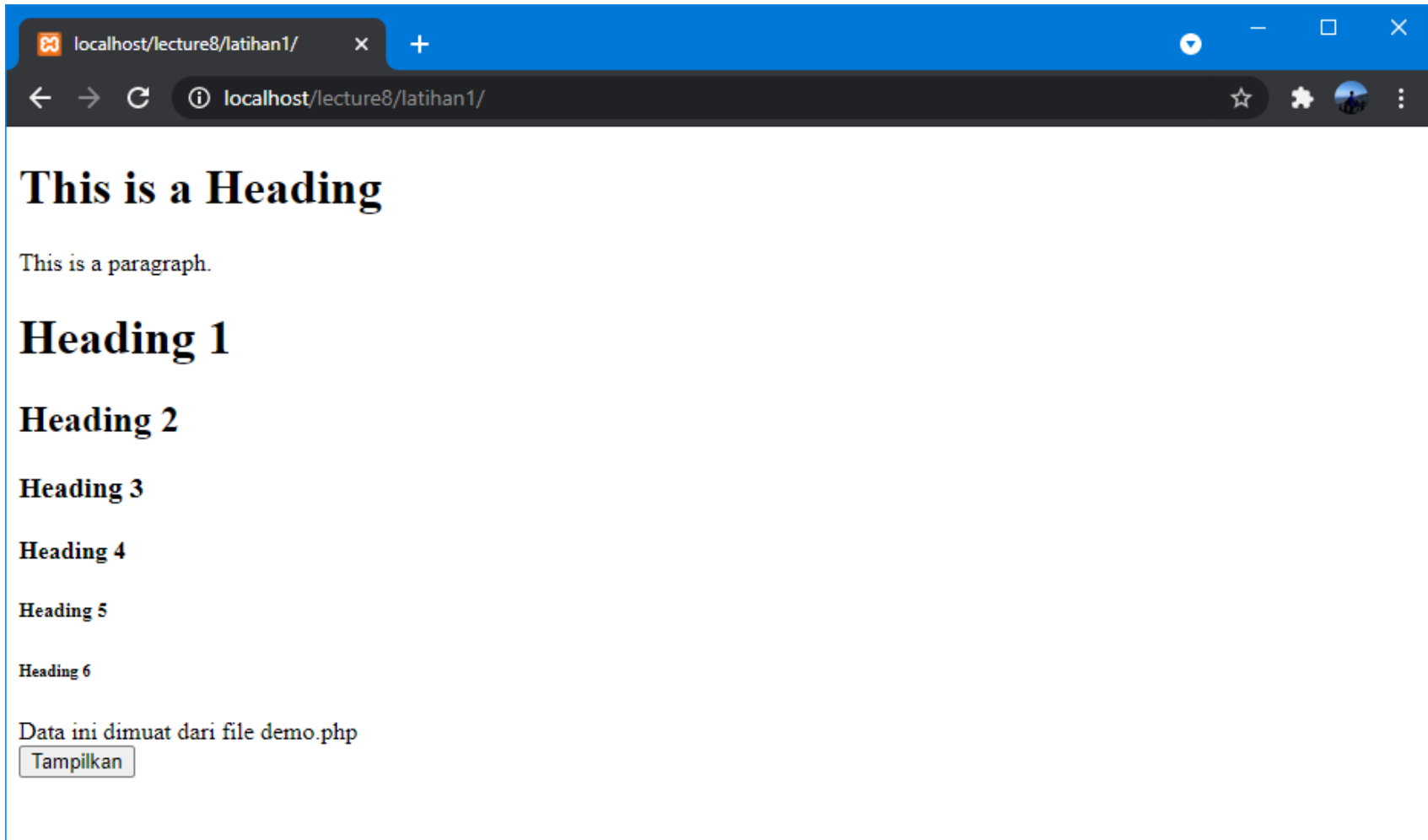
1 - Penggunaan Metode Load()

Langkah 2 : Buat file **index.php**

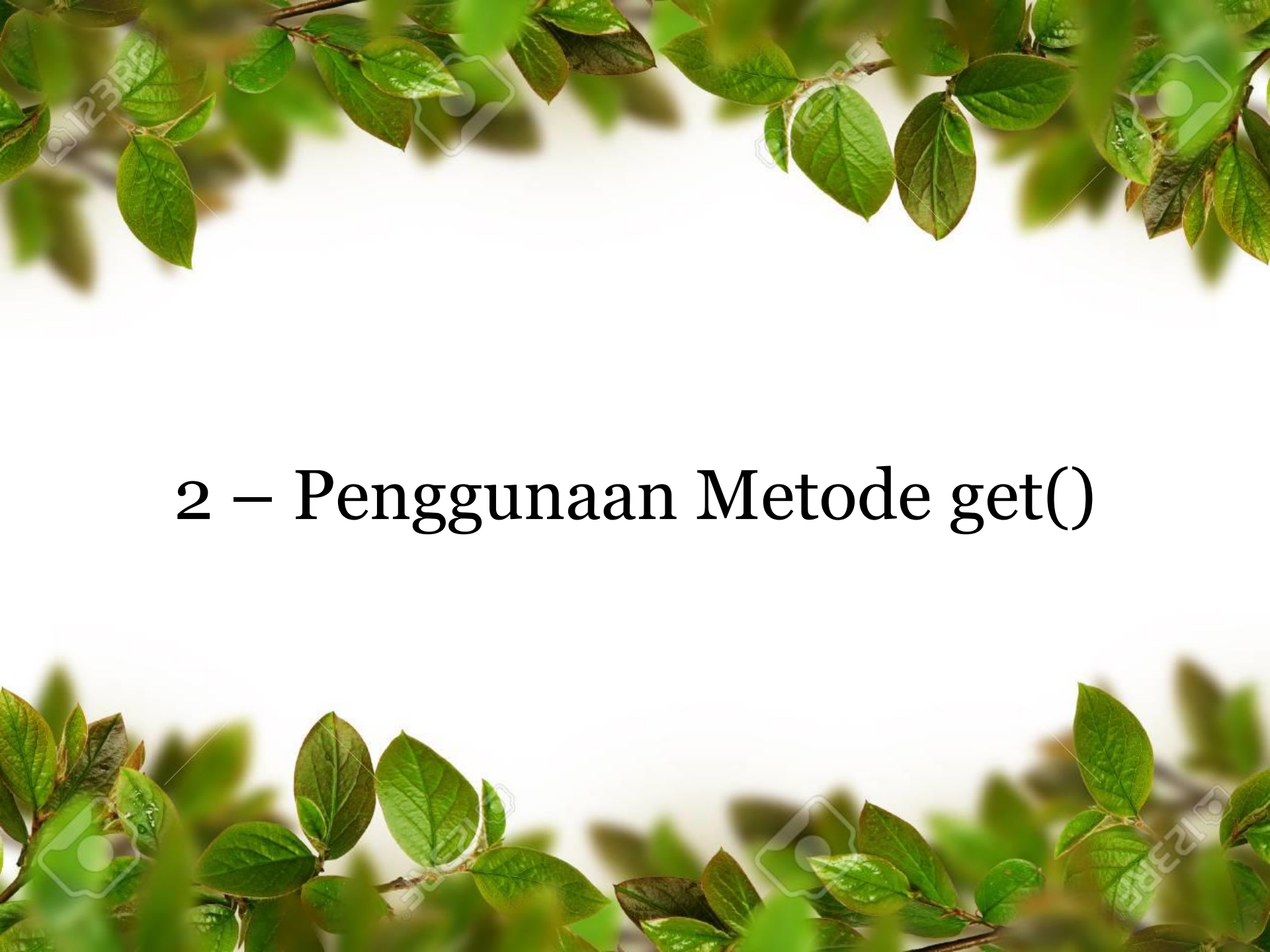
Perhatikan pada bagian `<head>` kita menyisipkan file ***jquery-3.5.1.min.js*** yang berada di dalam satu folder project di **htdocs**.



1 - Penggunaan Metode Load()



Elemen yang dipakai untuk menampilkan data yang di load adalah pada elemen `<div>` dengan atribut `id='tampil'`, Saat tombol dengan atribut `id='btn_tampil'` di klik maka isi dari file **demo.php** akan dimuat ke dalam elemen `id='tampil'`.



2 – Penggunaan Metode get()

2 - Metode jQuery get()

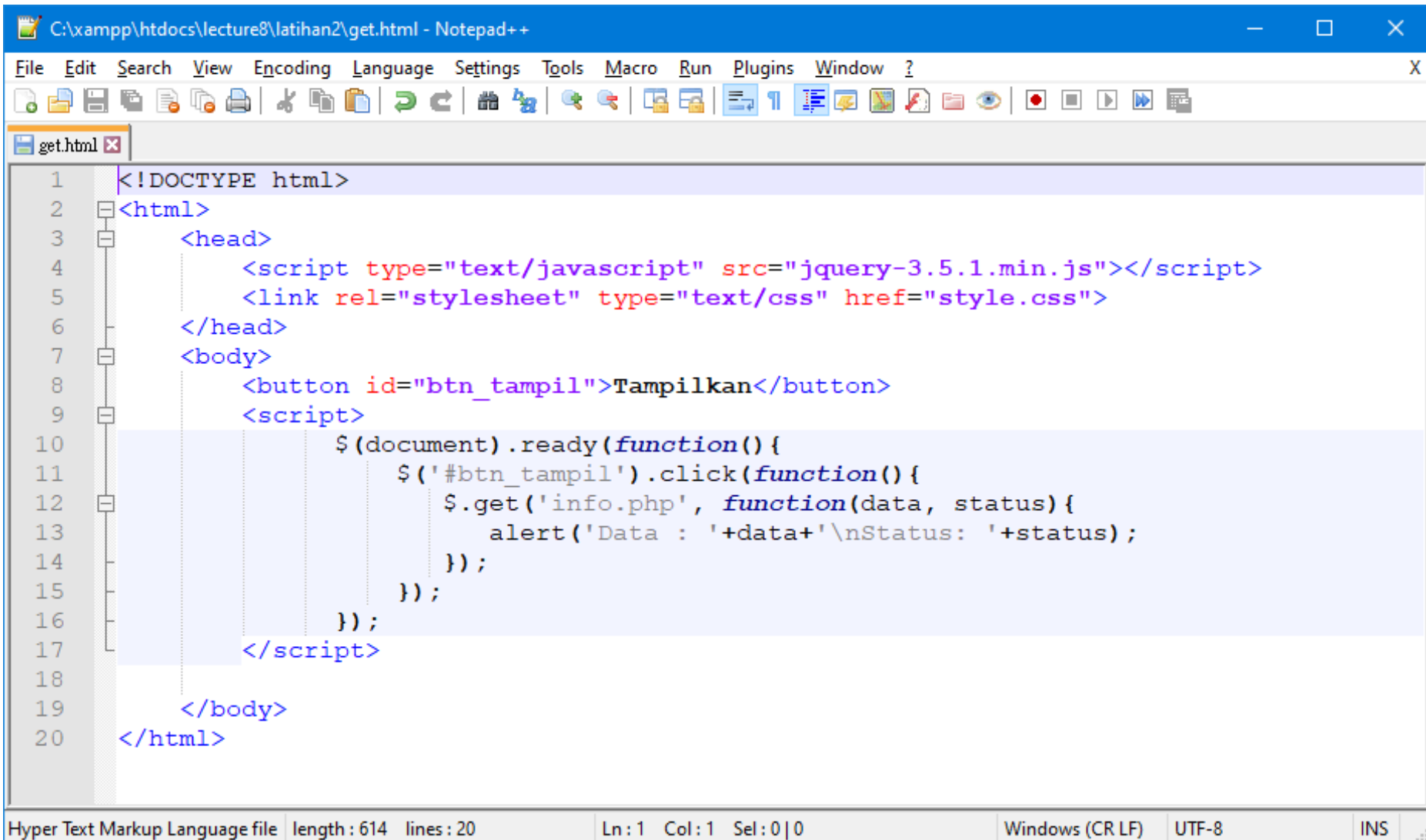
Metode get() akan meminta data dari server dengan jenis HTTP GET dengan bentuk umum sebagai berikut:

```
$.get (URL,callback);
```

Parameter URL digunakan untuk menentukan URL yang diminta, sedangkan parameter callback adalah fungsi yang akan dieksekusi saat permintaan berhasil.

2 - Metode jQuery get()

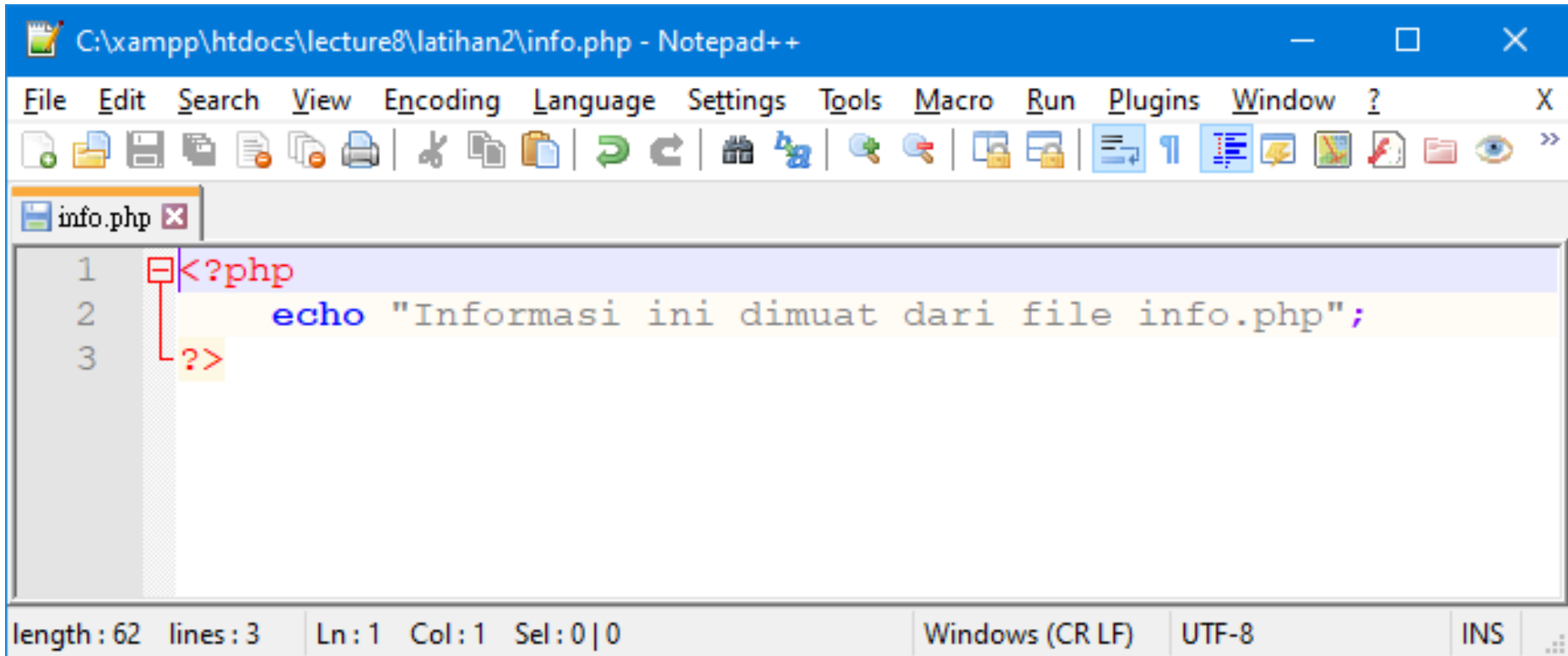
Langkah #1 : Buat file *get.html*



```
C:\xampp\htdocs\lecture8\latihan2\get.html - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
get.html x
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <script type="text/javascript" src="jquery-3.5.1.min.js"></script>
5     <link rel="stylesheet" type="text/css" href="style.css">
6   </head>
7   <body>
8     <button id="btn_tampil">Tampilkan</button>
9     <script>
10      $(document).ready(function() {
11        $('#btn_tampil').click(function() {
12          $.get('info.php', function(data, status) {
13            alert('Data : '+data+'\nStatus: '+status);
14          });
15        });
16      });
17    </script>
18  </body>
19 </html>
20
Hyper Text Markup Language file length : 614 lines : 20 Ln: 1 Col: 1 Sel: 0 | 0 Windows (CR LF) UTF-8 INS
```

2 - Metode jQuery get()

Langkah #2 : Buat file baru dengan nama *info.php*



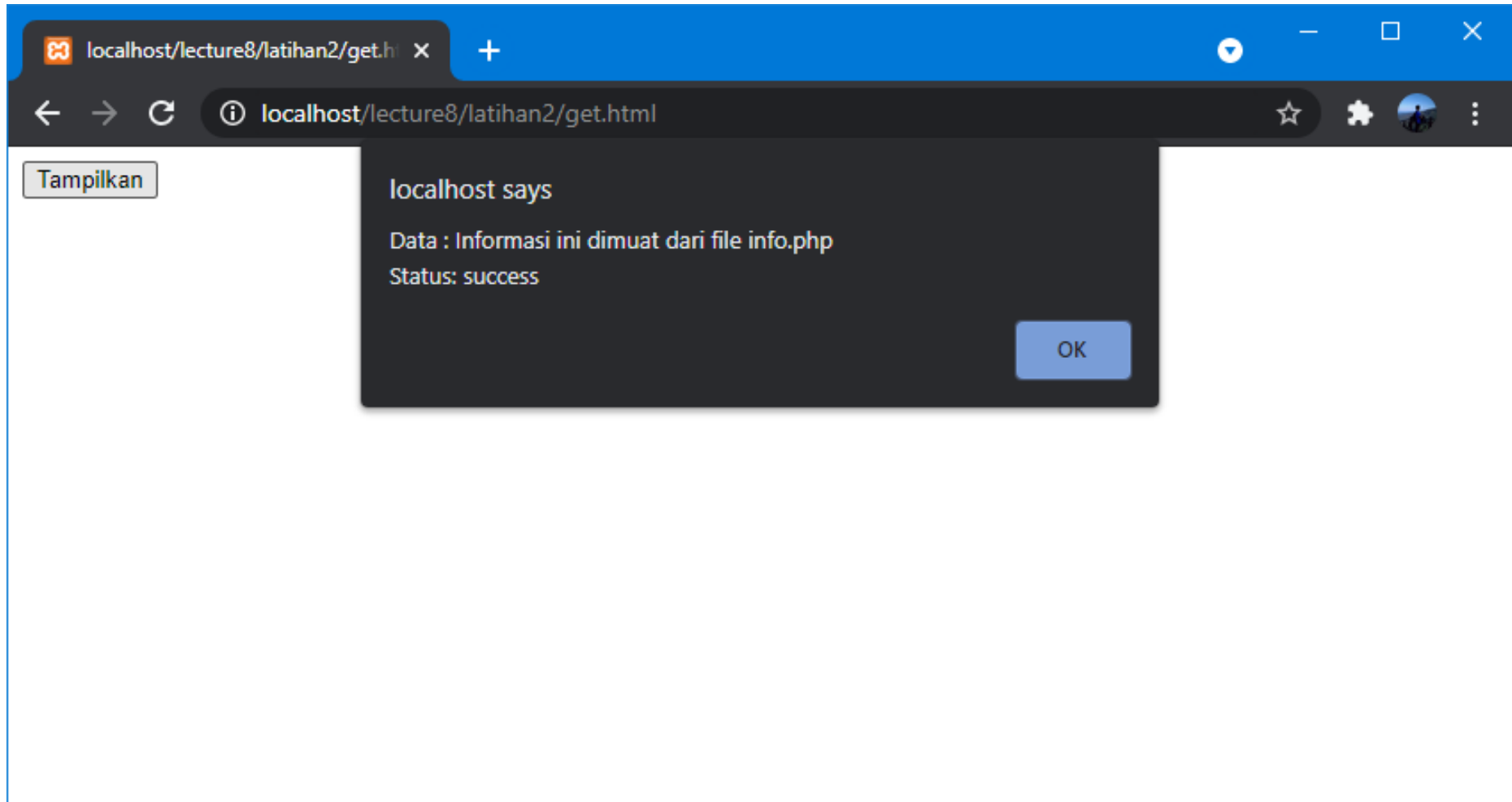
The screenshot shows the Notepad++ application window. The title bar reads "C:\xampp\htdocs\lecture8\latihan2\info.php - Notepad++". The menu bar includes File, Edit, Search, View, Encoding, Language, Settings, Tools, Macro, Run, Plugins, Window, and ?. The toolbar contains various icons for file operations and editing. The editor window has a tab for "info.php". The code is as follows:

```
1 <?php
2     echo "Informasi ini dimuat dari file info.php";
3 ?>
```

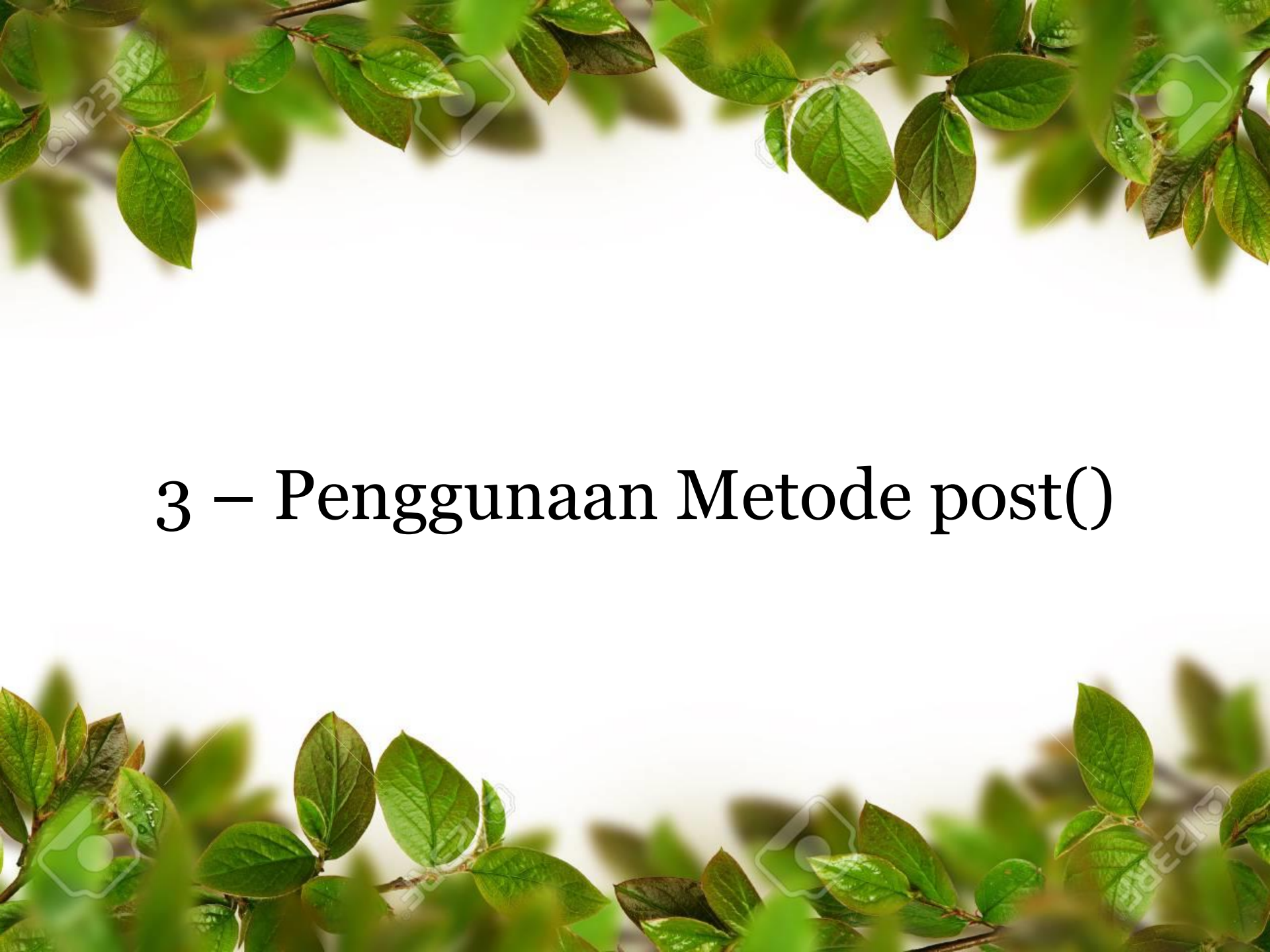
The status bar at the bottom displays "length : 62 lines : 3", "Ln : 1 Col : 1 Sel : 0 | 0", "Windows (CR LF)", "UTF-8", and "INS".

2 - Metode jQuery get()

Langkah #3 : Jalankan file *get.html* melalui localhost di *browser*



Saat **button** “Tampilkan” di klik maka **browser** akan menampilkan jendela pemberitahuan (**alert**) dengan menampilkan data yang diambil dari file eksternal PHP dan menampilkan status sukses.

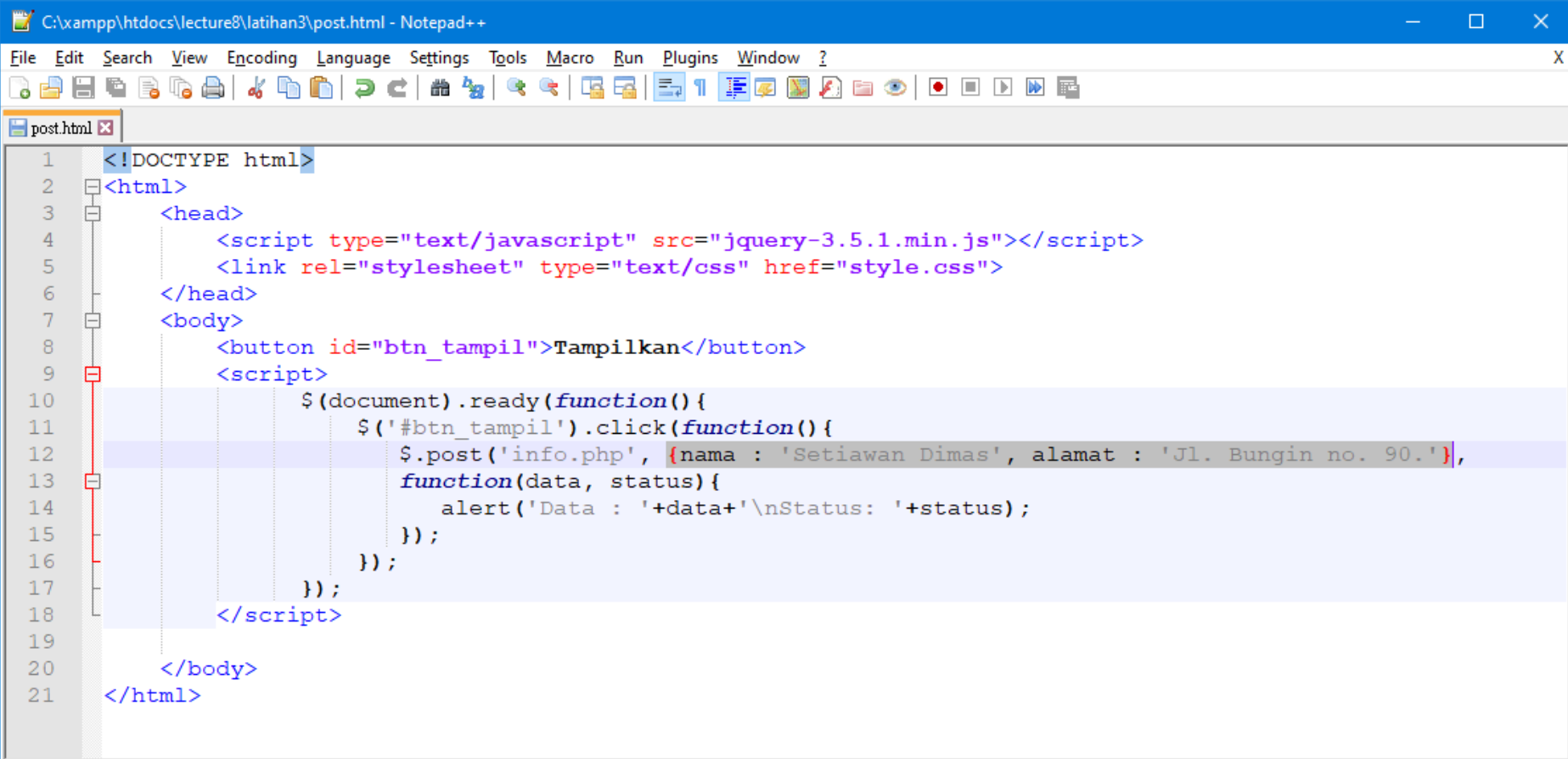


3 – Penggunaan Metode post()

3 - Metode jQuery post()

Pada metode *post*, terdapat sedikit tambahan pada *parameter opsional* dimana dapat menentukan beberapa data yang akan dikirim bersamaan saat permintaan (*request*) dilakukan.

Langkah #1 : Buat file *post.html*

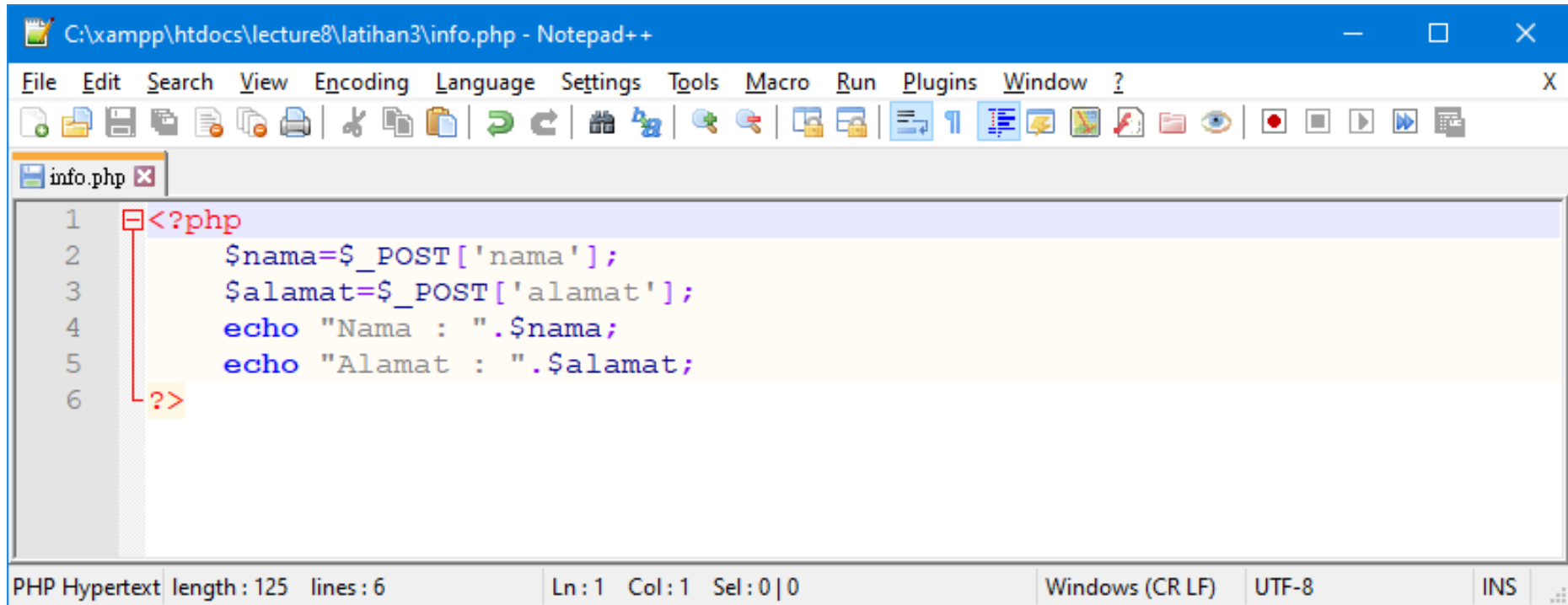


```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <script type="text/javascript" src="jquery-3.5.1.min.js"></script>
5     <link rel="stylesheet" type="text/css" href="style.css">
6   </head>
7   <body>
8     <button id="btn_tampil">Tampilkan</button>
9     <script>
10      $(document).ready(function() {
11        $('#btn_tampil').click(function() {
12          $.post('info.php', {nama : 'Setiawan Dimas', alamat : 'Jl. Bungin no. 90.'}),
13          function(data, status) {
14            alert('Data : '+data+'\nStatus: '+status);
15          });
16        });
17      });
18    </script>
19  </body>
20 </html>
```

Hyper Text Markup Language file length : 695 lines : 21 Ln : 12 Col : 97 Sel : 56 | 1 Windows (CR LF) UTF-8 INS

3 - Metode jQuery post()

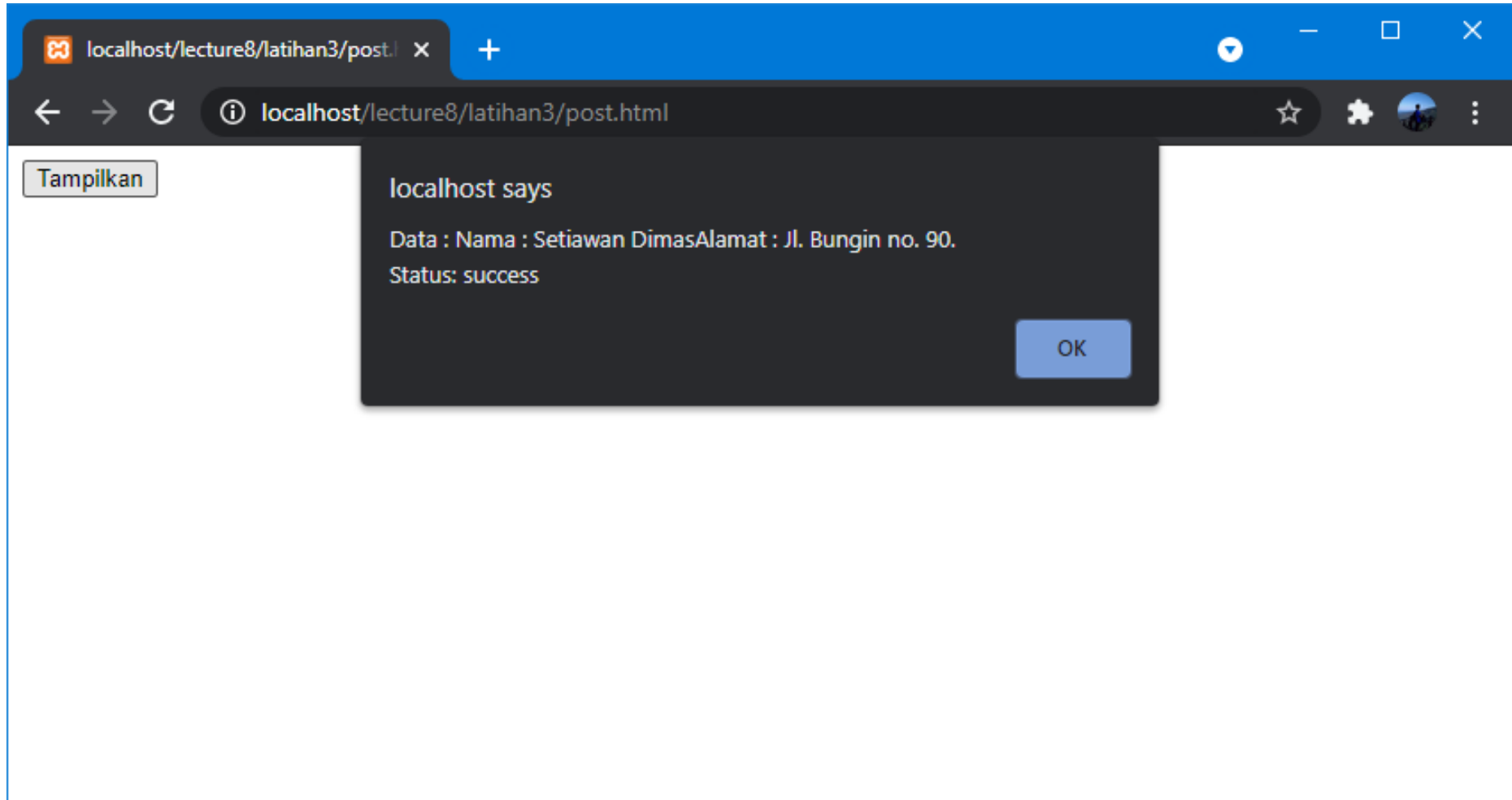
Langkah #2 : Buat file *info.php*



```
C:\xampp\htdocs\lecture8\latihan3\info.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
info.php
1 <?php
2     $nama=$_POST['nama'];
3     $alamat=$_POST['alamat'];
4     echo "Nama : ".$nama;
5     echo "Alamat : ".$alamat;
6 ?>
PHP Hypertext length : 125 lines : 6 Ln : 1 Col : 1 Sel : 0 | 0 Windows (CR LF) UTF-8 INS
```


3 - Metode jQuery post()

Langkah #2 : Jalankan file *post.html* melalui *localhost* di *web browser*



A decorative border of green leaves and branches at the top of the slide.

More Interactive Applications

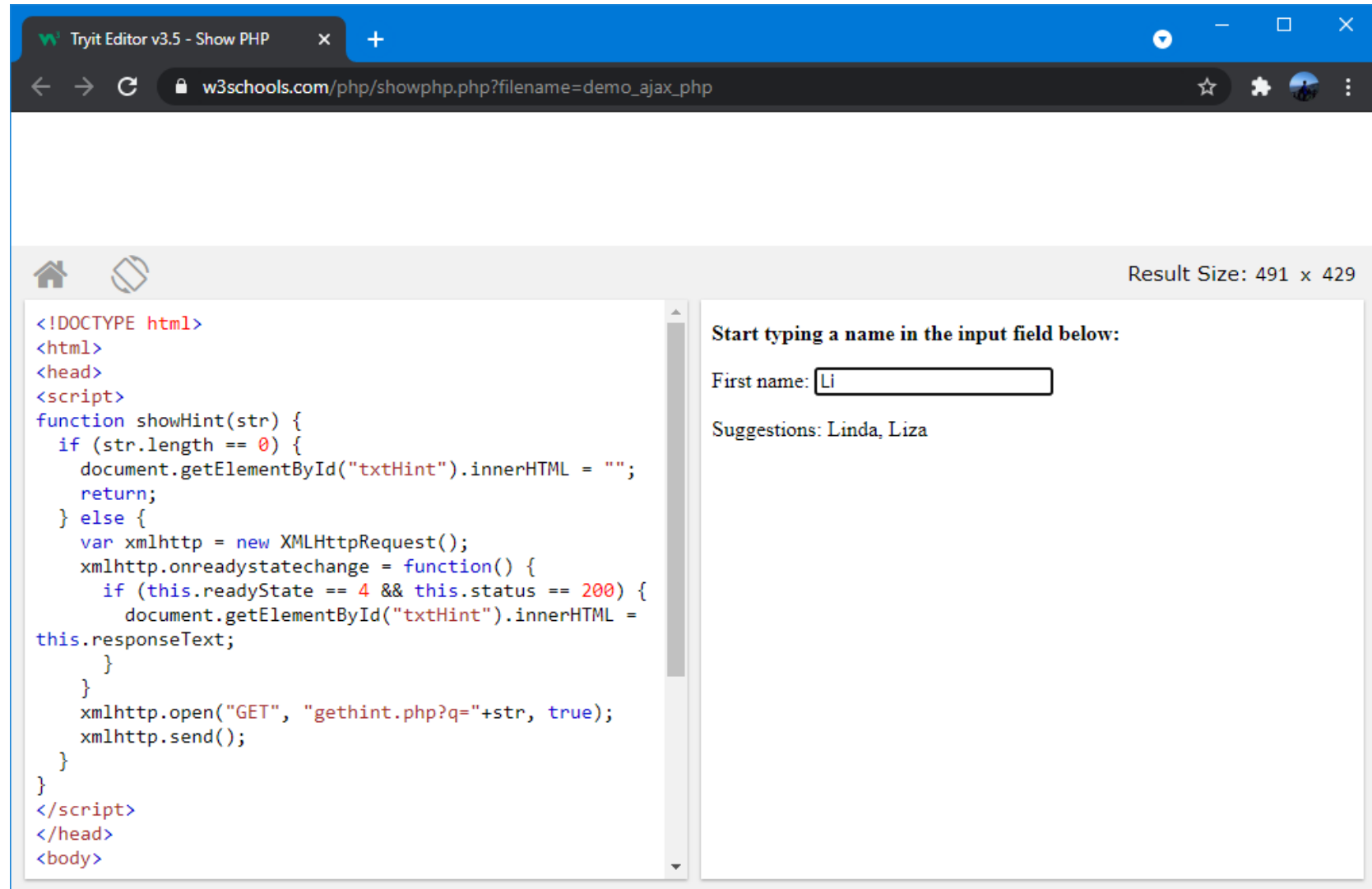
[AJAX and PHP]

A decorative border of green leaves and branches at the bottom of the slide.

More Interactive Applications

Simple Autocomplete Suggestion

Link: https://www.w3schools.com/php/php_ajax_php.asp



The screenshot shows a web browser window with the address bar displaying `w3schools.com/php/showphp.php?filename=demo_ajax_php`. The page content is divided into two main sections. On the left, there is a code editor showing the following PHP code:

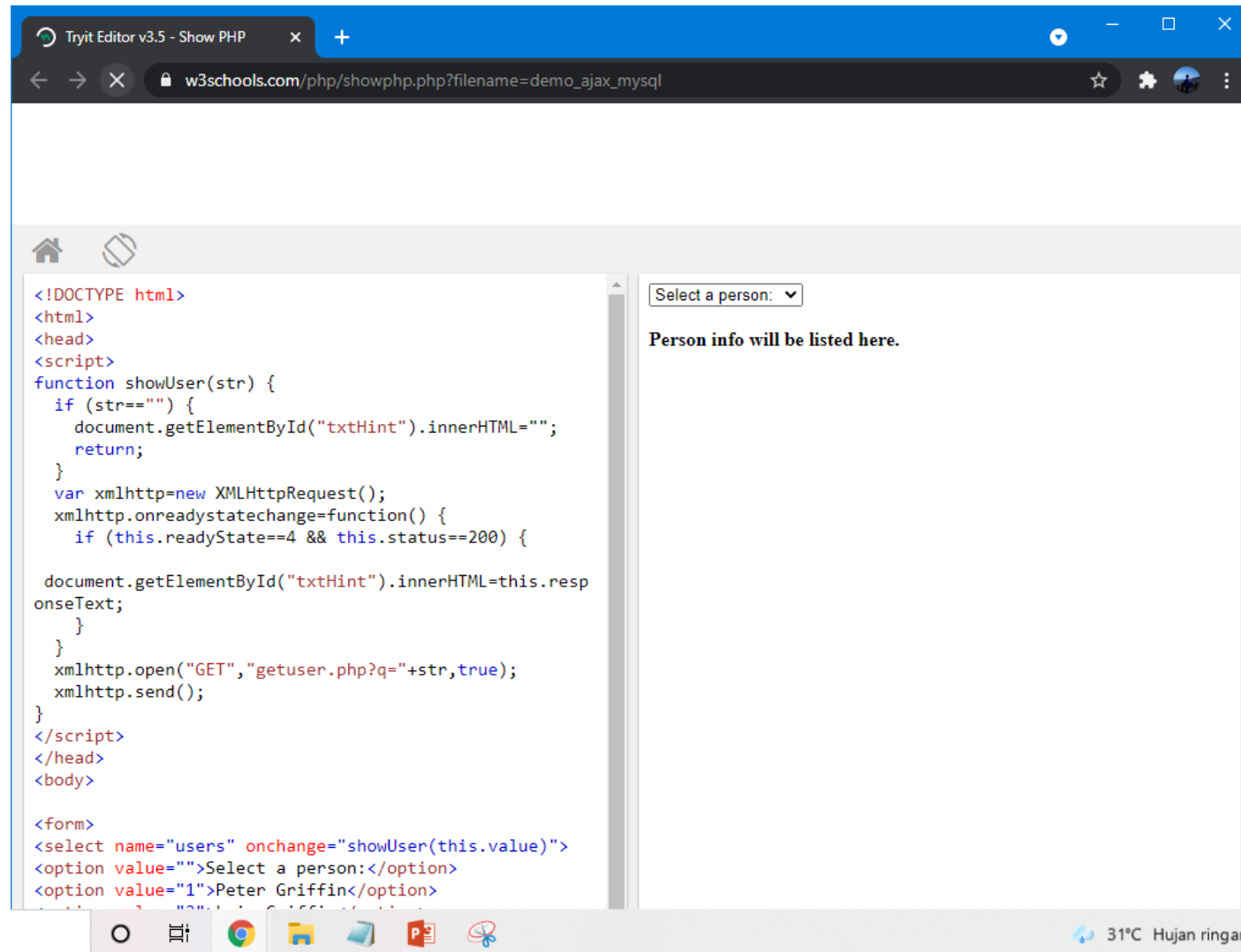
```
<!DOCTYPE html>
<html>
<head>
<script>
function showHint(str) {
  if (str.length == 0) {
    document.getElementById("txtHint").innerHTML = "";
    return;
  } else {
    var xmlhttp = new XMLHttpRequest();
    xmlhttp.onreadystatechange = function() {
      if (this.readyState == 4 && this.status == 200) {
        document.getElementById("txtHint").innerHTML =
this.responseText;
      }
    }
    xmlhttp.open("GET", "gethint.php?q="+str, true);
    xmlhttp.send();
  }
}
</script>
</head>
<body>
```

On the right, the browser displays the output of the script. It starts with the text "Start typing a name in the input field below:". Below this, there is a text input field with the value "Li". Underneath the input field, the text "Suggestions: Linda, Liza" is displayed. In the top right corner of the browser window, the text "Result Size: 491 x 429" is visible.

More Interactive Applications

PHP - AJAX and MySQL (AJAX Database Example)

Link: https://www.w3schools.com/php/php_ajax_database.asp



The screenshot shows a web browser window with the URL `w3schools.com/php/showphp.php?filename=demo_ajax_mysql`. The page displays a dropdown menu with the text "Select a person: v" and a text area below it that says "Person info will be listed here." The left sidebar shows the source code of the page, which is a PHP script using AJAX to fetch user data from a database.

```
<!DOCTYPE html>
<html>
<head>
<script>
function showUser(str) {
  if (str=="") {
    document.getElementById("txtHint").innerHTML="";
    return;
  }
  var xmlhttp=new XMLHttpRequest();
  xmlhttp.onreadystatechange=function() {
    if (this.readyState==4 && this.status==200) {

document.getElementById("txtHint").innerHTML=this.respon
seText;
    }
  }
  xmlhttp.open("GET","getuser.php?q="+str,true);
  xmlhttp.send();
}
</script>
</head>
<body>

<form>
<select name="users" onchange="showUser(this.value)">
<option value="">Select a person:</option>
<option value="1">Peter Griffin</option>
</select>
</form>
```

More Interactive Applications

PHP Example - AJAX Live Search

Link: https://www.w3schools.com/php/php_ajax_livesearch.asp

AJAX Live Search

The following example will demonstrate a live search, where you get search results while you type.

Live search has many benefits compared to traditional searching:

- Results are shown as you type
- Results narrow as you continue typing
- If results become too narrow, remove characters to see a broader result

Search for a W3Schools page in the input field below:

[CSS background](#)

[Property](#)

[CSS border Property](#)

[JavaScript Date Object](#)

[JavaScript Array Object](#)

More Interactive Applications

PHP Example - AJAX Poll

Link: https://www.w3schools.com/php/php_ajax_poll.asp

AJAX Poll

The following example will demonstrate a poll where the result is shown without reloading.

Result:

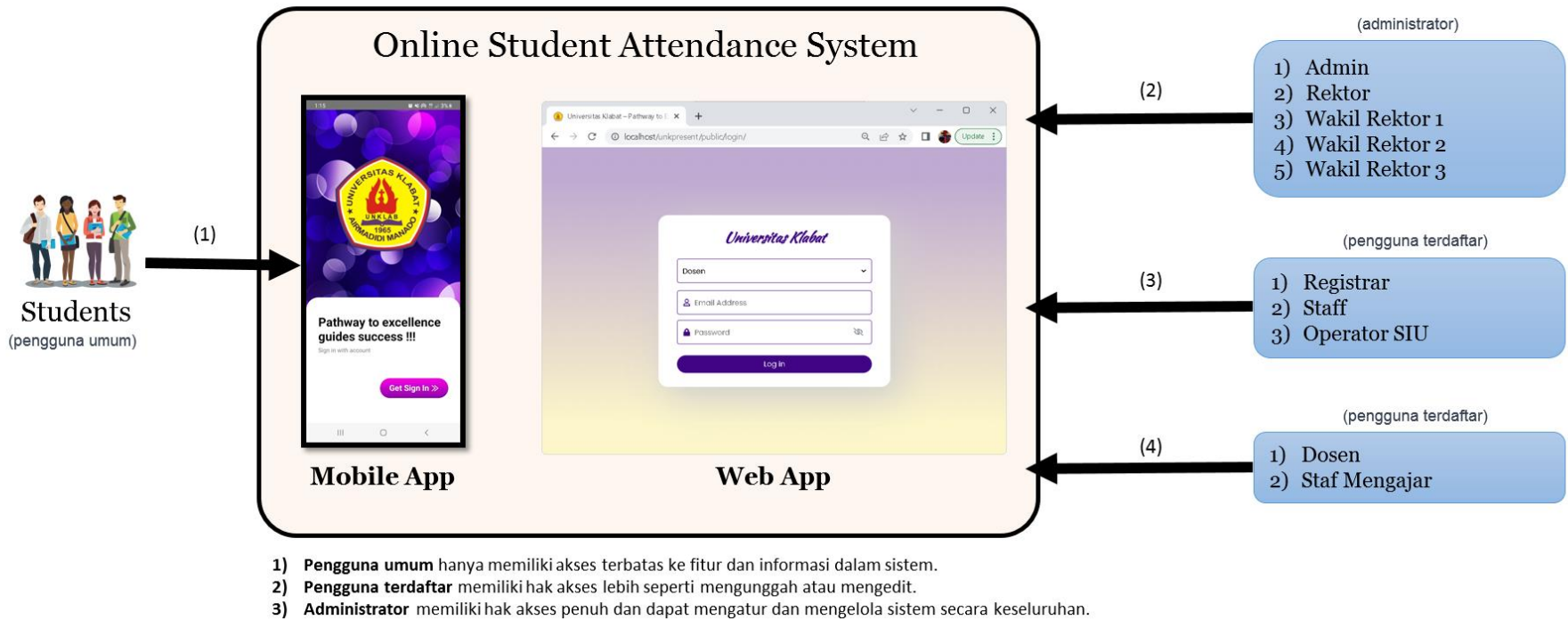
Yes:  79%

No:  21%

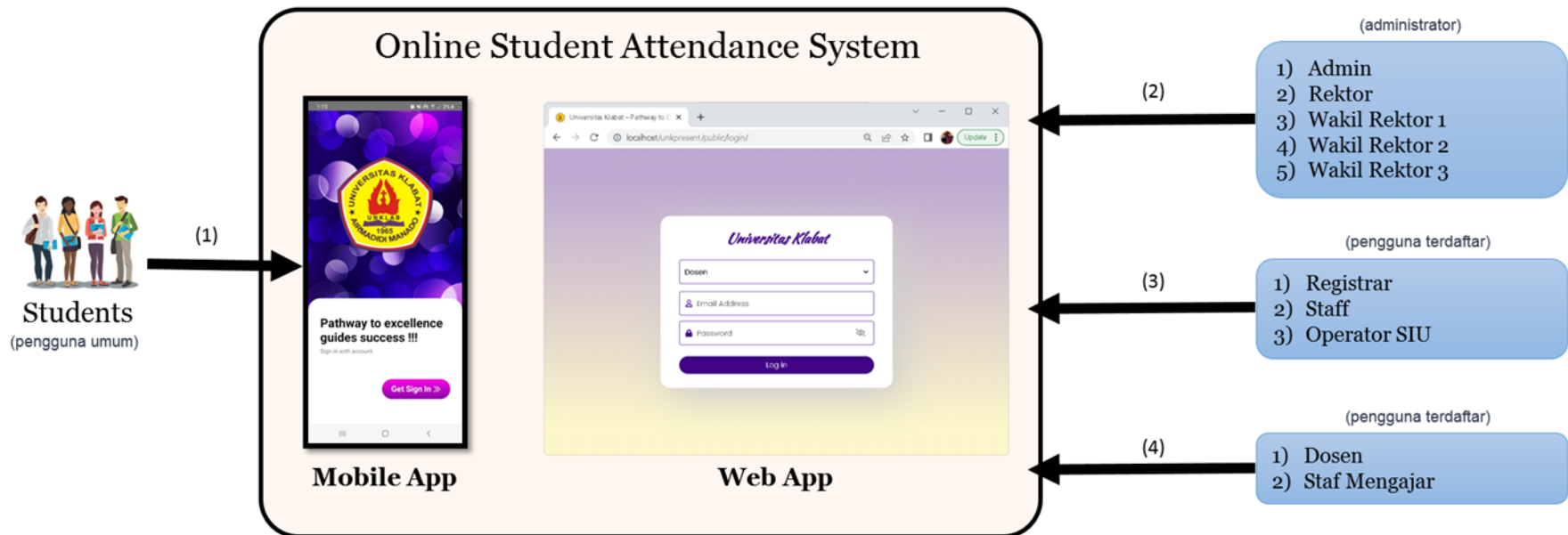
A decorative border of green leaves and branches at the top of the page.

How to use **ajax()** Method?

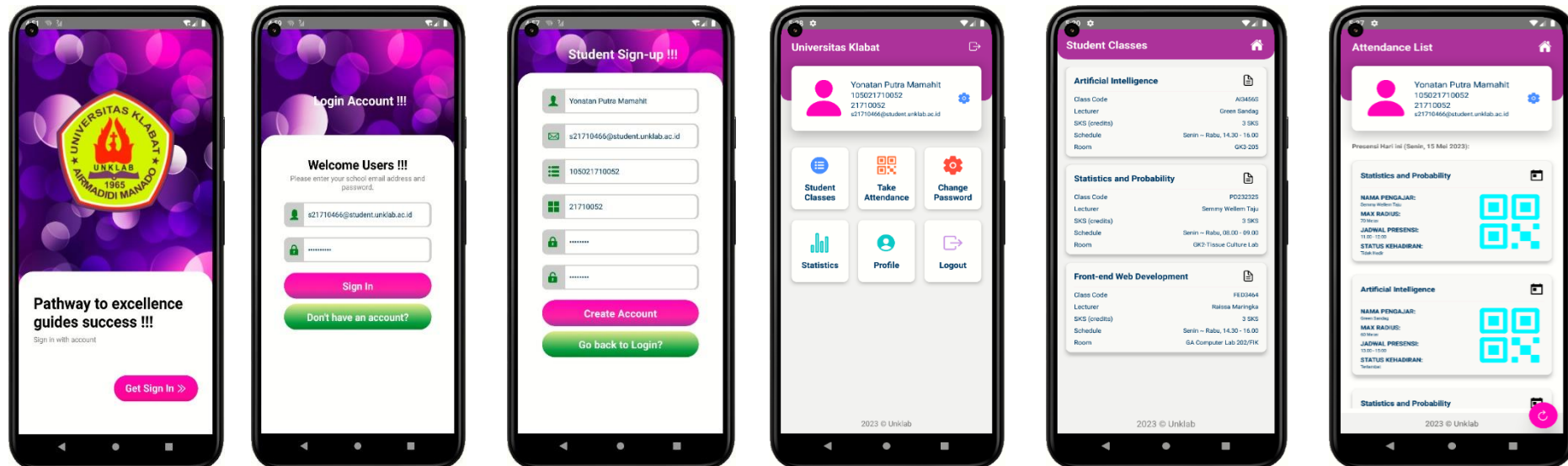
A decorative border of green leaves and branches at the bottom of the page.

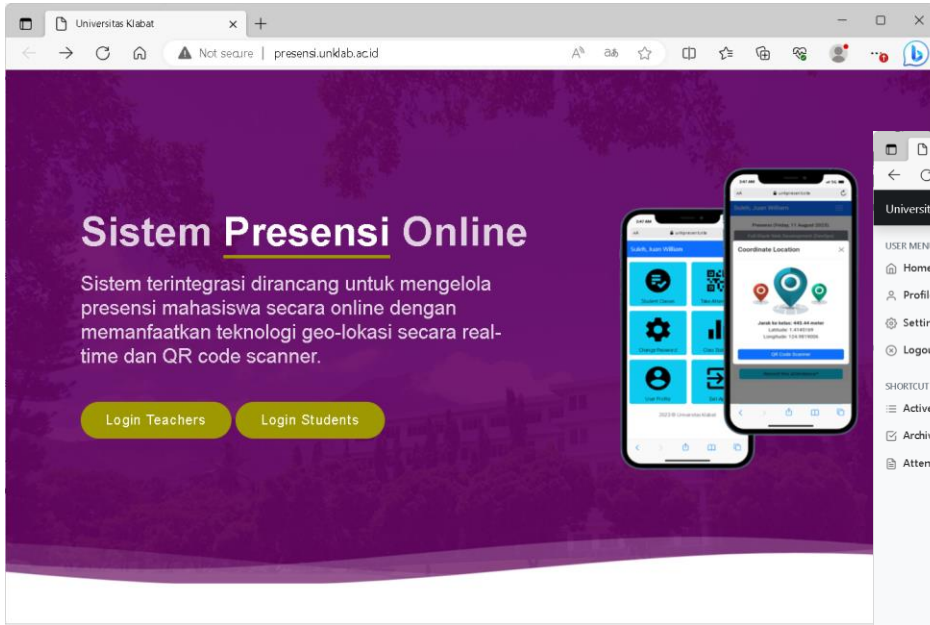


Sistem absensi mahasiswa secara online dengan dua aplikasi (*mobile app* dan *web app*) memiliki tujuan untuk memudahkan *students* dan *staf* dalam melakukan absensi dan pengelolaan kehadiran mahasiswa di kelas.

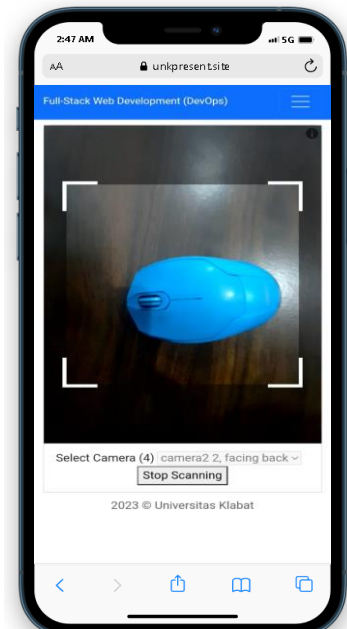
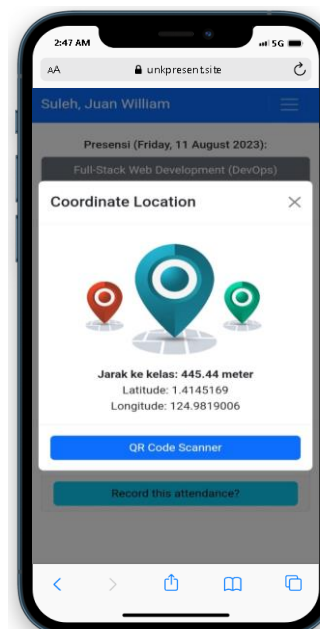
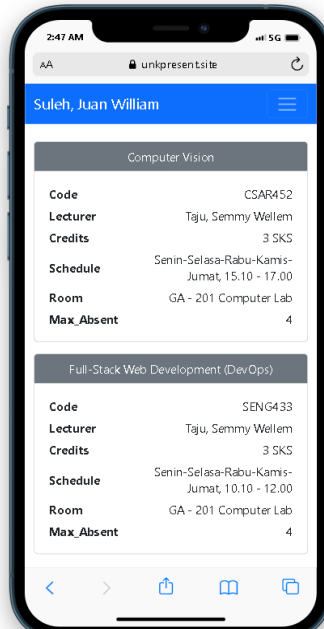
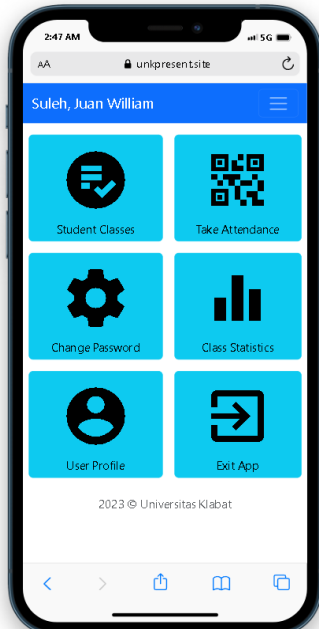
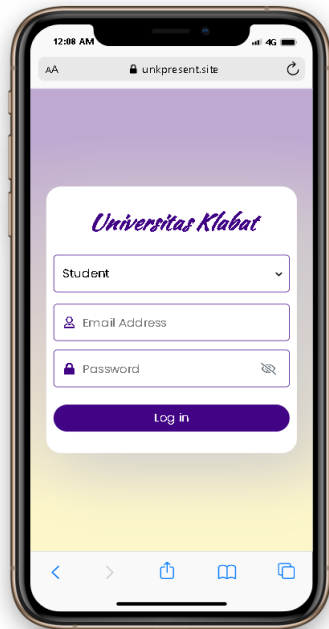


- 1) **Pengguna umum** hanya memiliki akses terbatas ke fitur dan informasi dalam sistem.
- 2) **Pengguna terdaftar** memiliki hak akses lebih seperti mengunggah atau mengedit.
- 3) **Administrator** memiliki hak akses penuh dan dapat mengatur dan mengelola sistem secara keseluruhan.





No.	Students	Number meetings?	Hadir	Late	Izin	Alpa
1	Abram, Mogandi Timotius S2200060@student.unklab.ac.id	19 Meetings	19 times	0 times	0 times	0 times
2	Amping, Bryan S2200047@student.unklab.ac.id	19 Meetings	18 times	1 times	0 times	0 times
3	Dumais, Ade Pricilia S2200153@student.unklab.ac.id	19 Meetings	19 times	0 times	0 times	0 times
4	Haerani, Christian Solafide S2200045@student.unklab.ac.id	19 Meetings	14 times	3 times	0 times	2 times
5	Kabo, Deo Timothy S2200037@student.unklab.ac.id	19 Meetings	16 times	2 times	1 times	0 times
6	Korengkeng, Vito Julio S2200432@student.unklab.ac.id	19 Meetings	18 times	1 times	0 times	0 times
7	Mumu, Griffin Meidy S2200054@student.unklab.ac.id	19 Meetings	18 times	0 times	0 times	1 times
	Mahut, Drivan Doter					



Metode jQuery ajax()

Ajax POST request with jQuery and PHP

```
1 $.ajax({
2   type: "POST",
3   url: 'script.php',
4   data: {name: 'John'},
5   success: function(data){
6     console.log(data);
7   },
8   error: function(xhr, status, error){
9     console.error(xhr);
10  }
11 });
```

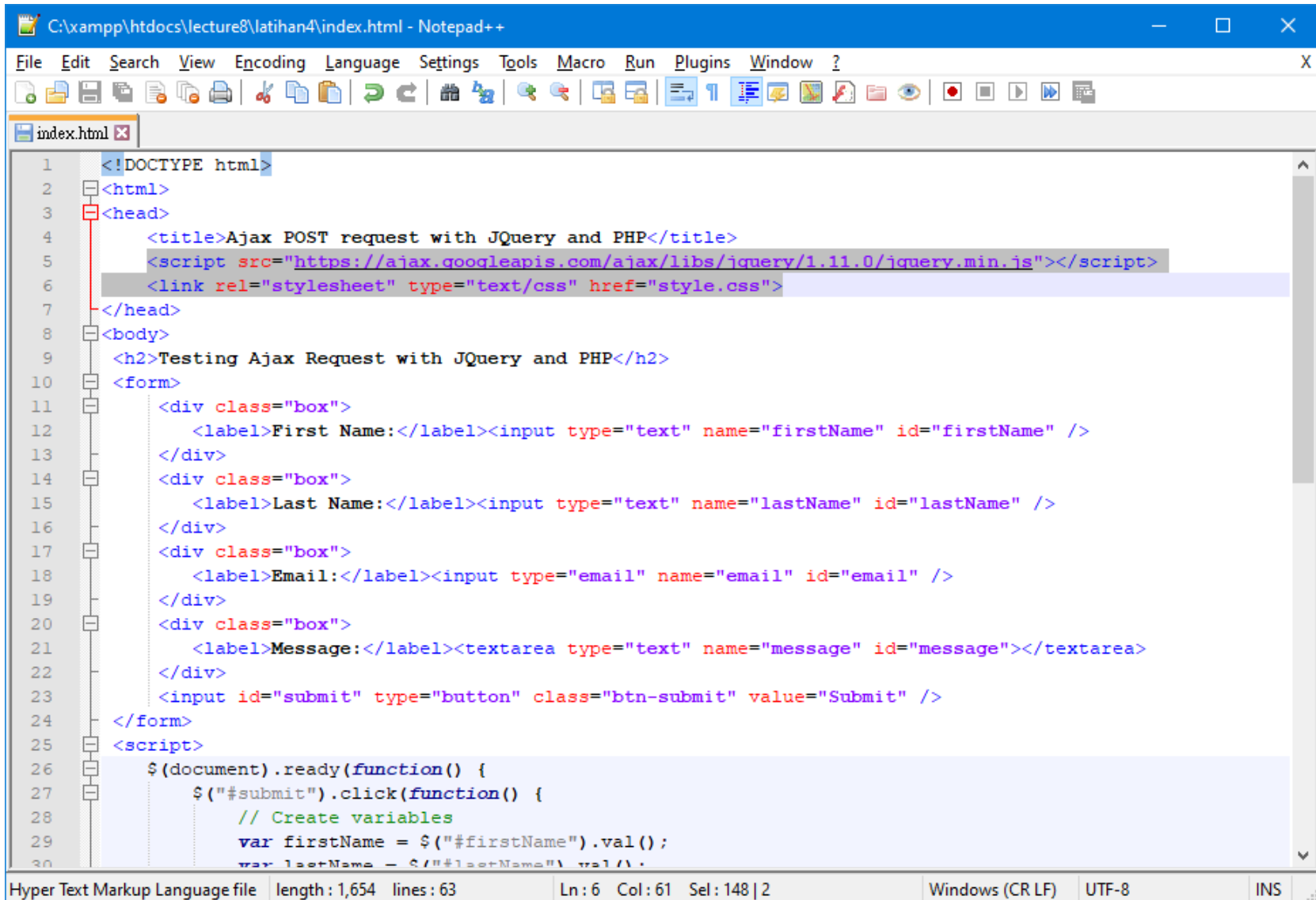
Basic Ajax POST request

Ajax Parameters:

- ❑ **type** – Type of http request like POST, GET, PUT, DELETE.
- ❑ **url** – The external URL that we want to send the ajax request. Here, we used ***script.php***.
- ❑ **data** – Data which we want to send in the request. We have passed {name: 'John'} in the above code.
- ❑ **success** – If the ajax request runs successfully then this function is called. In the above function we have a data parameter that is the output of the request url.
- ❑ **error** – If the ajax request failed then this function is called so we can debug the issue.

Metode jQuery ajax()

Langkah #1 : Create an HTML form in file *index.html*

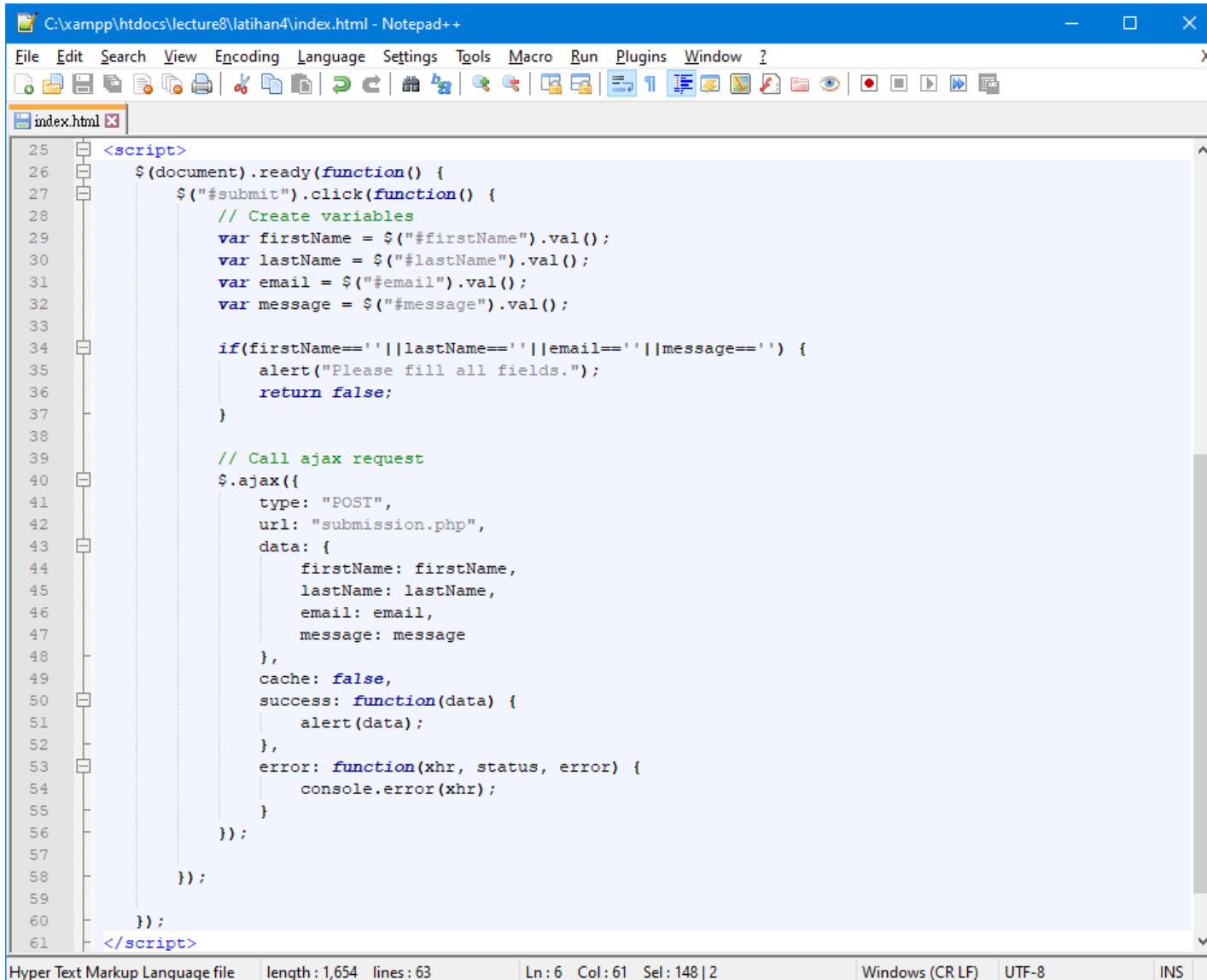


```
C:\xampp\htdocs\lecture8\latihan4\index.html - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
index.html x
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <title>Ajax POST request with JQuery and PHP</title>
5     <script src="https://ajax.googleapis.com/ajax/libs/jquery/1.11.0/jquery.min.js"></script>
6     <link rel="stylesheet" type="text/css" href="style.css">
7 </head>
8 <body>
9     <h2>Testing Ajax Request with JQuery and PHP</h2>
10    <form>
11        <div class="box">
12            <label>First Name:</label><input type="text" name="firstName" id="firstName" />
13        </div>
14        <div class="box">
15            <label>Last Name:</label><input type="text" name="lastName" id="lastName" />
16        </div>
17        <div class="box">
18            <label>Email:</label><input type="email" name="email" id="email" />
19        </div>
20        <div class="box">
21            <label>Message:</label><textarea type="text" name="message" id="message"></textarea>
22        </div>
23        <input id="submit" type="button" class="btn-submit" value="Submit" />
24    </form>
25    <script>
26        $(document).ready(function() {
27            $("#submit").click(function() {
28                // Create variables
29                var firstName = $("#firstName").val();
30                var lastName = $("#lastName").val();
```

Hyper Text Markup Language file | length: 1,654 | lines: 63 | Ln: 6 Col: 61 Sel: 148 | 2 | Windows (CR LF) | UTF-8 | INS

Metode jQuery ajax()

Langkah #1 : Create an HTML form in file *index.html*

A screenshot of a Notepad++ text editor window. The title bar shows the file path: C:\xampp\htdocs\lecture8\latihan4\index.html - Notepad++. The menu bar includes File, Edit, Search, View, Encoding, Language, Settings, Tools, Macro, Run, Plugins, Window, and Help. The toolbar contains various icons for file operations and editing. The editor window shows the file index.html with a list of line numbers on the left. The code is as follows:

```
25 <script>
26 $(document).ready(function() {
27     $("#submit").click(function() {
28         // Create variables
29         var firstName = $("#firstName").val();
30         var lastName = $("#lastName").val();
31         var email = $("#email").val();
32         var message = $("#message").val();
33
34         if(firstName==''||lastName==''||email==''||message=='') {
35             alert("Please fill all fields.");
36             return false;
37         }
38
39         // Call ajax request
40         $.ajax({
41             type: "POST",
42             url: "submission.php",
43             data: {
44                 firstName: firstName,
45                 lastName: lastName,
46                 email: email,
47                 message: message
48             },
49             cache: false,
50             success: function(data) {
51                 alert(data);
52             },
53             error: function(xhr, status, error) {
54                 console.error(xhr);
55             }
56         });
57     });
58 });
59
60
61 </script>
```

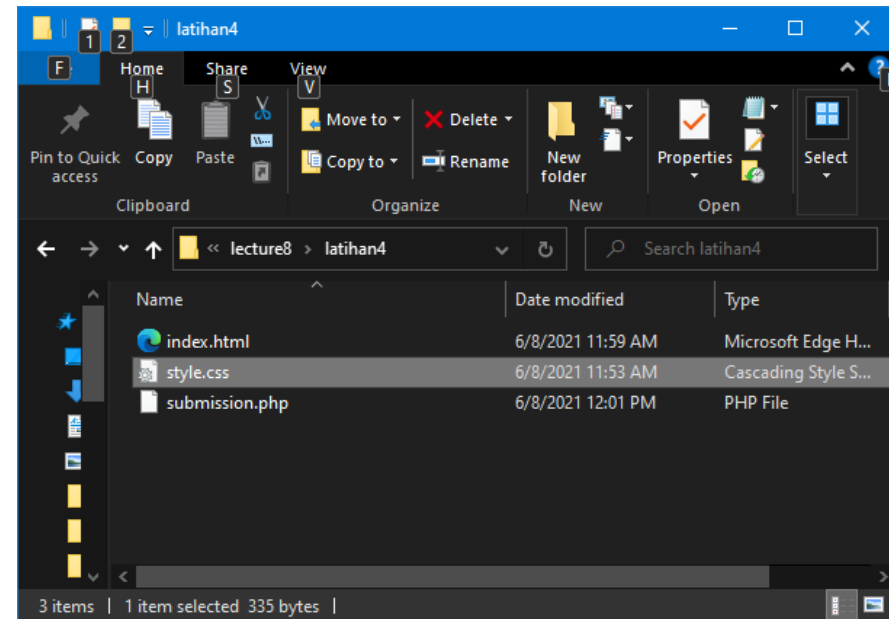
The status bar at the bottom indicates: Hyper Text Markup Language file, length: 1,654 lines: 63, Ln: 6 Col: 61 Sel: 148 | 2, Windows (CR LF), UTF-8, and INS.

Metode jQuery ajax()

Langkah #1 : Create an HTML form in file *index.html*

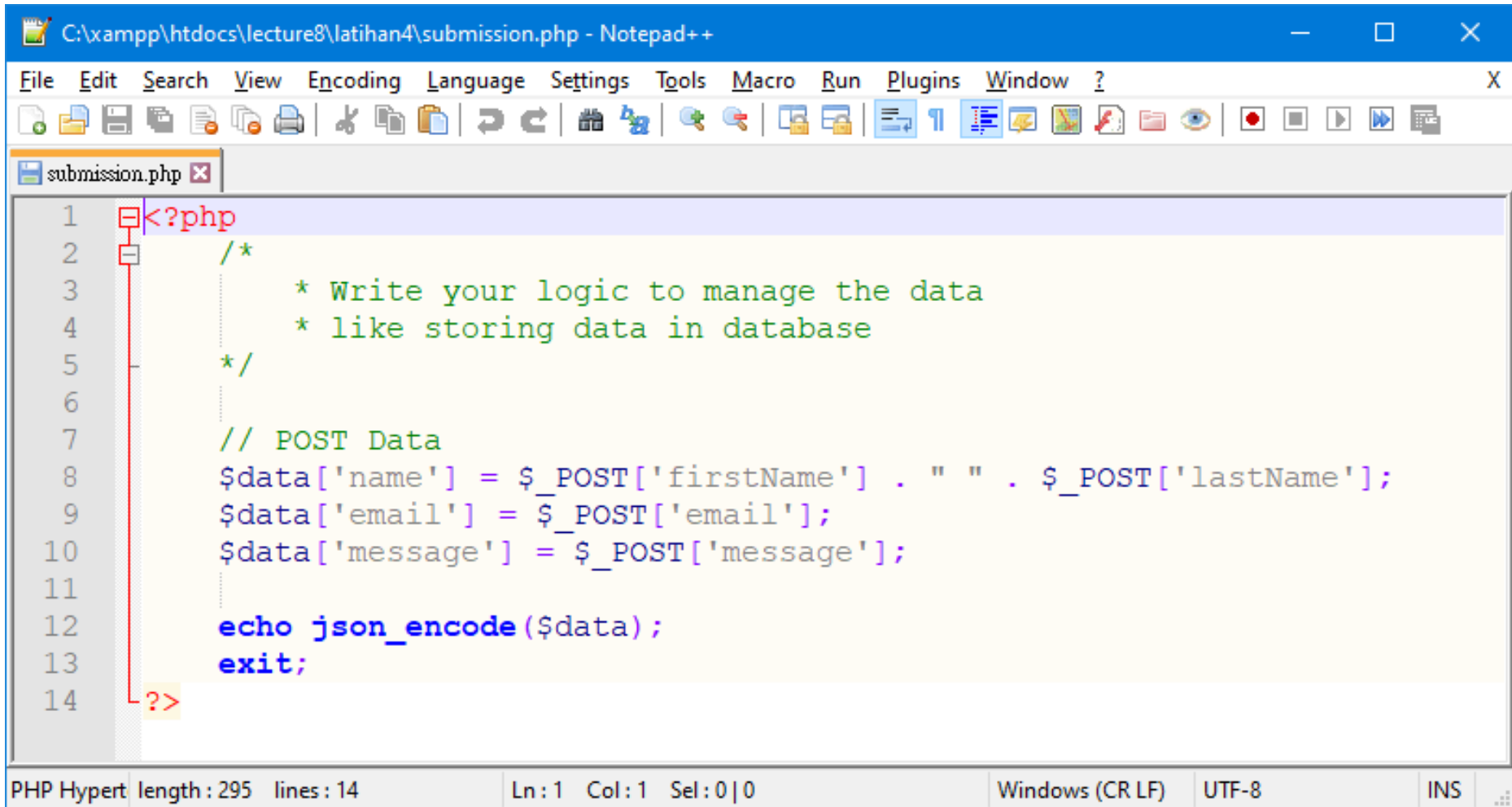
Create .CSS file to make a simple style on your webpage.

```
C:\xampp\htdocs\lecture8\latihan4\style.css - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run
Plugins Window ?
style.css
1 body {
2   font-family: calibri;
3 }
4
5 .box {
6   margin-bottom: 10px;
7 }
8
9 .box label {
10  display: inline-block;
11  width: 80px;
12  text-align: right;
13  margin-right: 10px;
14 }
15
16 .box input, .box textarea {
17  border-radius: 3px;
18  border: 1px solid #ddd;
19  padding: 5px 10px;
20 }
21
22 .btn-submit {
23   margin-left: 90px;
24 }
```



Metode jQuery ajax()

Langkah #2 : Create file ***submission.php*** to handle POST data in PHP file.

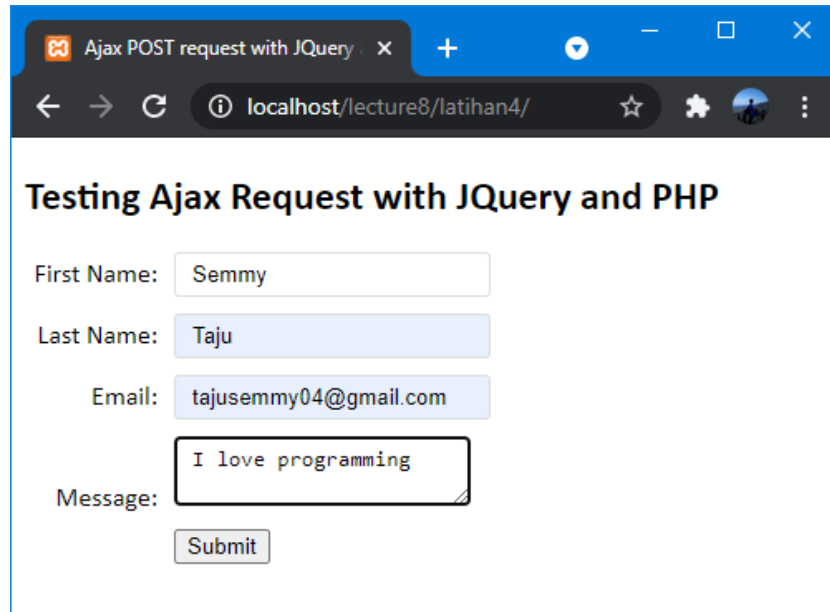


```
C:\xampp\htdocs\lecture8\latihan4\submission.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
submission.php
1 <?php
2 /*
3      * Write your logic to manage the data
4      * like storing data in database
5 */
6
7 // POST Data
8 $data['name'] = $_POST['firstName'] . " " . $_POST['lastName'];
9 $data['email'] = $_POST['email'];
10 $data['message'] = $_POST['message'];
11
12 echo json_encode($data);
13 exit;
14 ?>
```

PHP Hypert length : 295 lines : 14 Ln : 1 Col : 1 Sel : 0 | 0 Windows (CR LF) UTF-8 INS

Metode jQuery ajax()

Langkah #3 : Run *index.html* and check the output in the browser.



Ajax POST request with JQuery x

localhost/lecture8/latihan4/

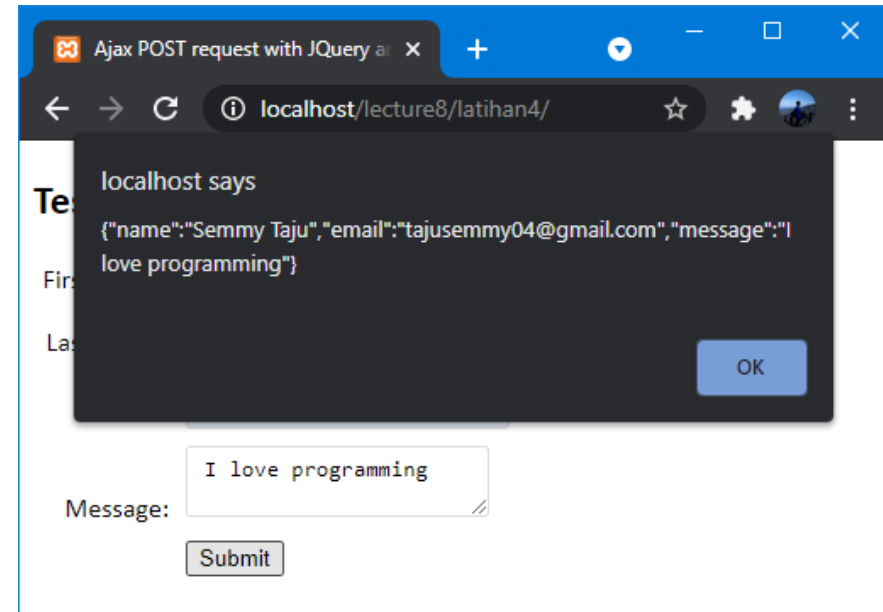
Testing Ajax Request with JQuery and PHP

First Name:

Last Name:

Email:

Message:



Ajax POST request with JQuery a x

localhost/lecture8/latihan4/

localhost says

```
{"name":"Semmy Taju","email":"tajusemmy04@gmail.com","message":"I love programming"}
```

OK

Message:

Tambahkan input data baru yaitu tanggal lahir dengan menggunakan HTML date picker (HTML `<input type="date">`)?

Show a Date Control

BirthDay:

September 2023

Su	Mo	Tu	We	Th	Fr	Sa
27	28	29	30	31	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
1	2	3	4	5	6	7

Clear Today

Attendance

Universitas Klabat

Search

Sign out

USER MENU

- Home
- Profile
- Settings
- Logout

SHORTCUT LECTURER

- Active Classes
- Archived Classes
- Attendance
- Magic Cut

Attendance Schedule

Course Topic:

Course:

Meeting:

Lecturer Email:

Date:

Time:

Max Radius:

September 2023

Su	Mo	Tu	We	Th	Fr	Sa
27	28	29	30	31	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
1	2	3	4	5	6	7

Summary

Back

Radius	Actions
50 Meter	Options
50 Meter	Options
100 Meter	Options
70 Meter	Options
50 Meter	Options

Taju, Semmy Wellem
semmy@unklab.ac.id

22 August 2023
08.30 - 10.00

11/11 Students

Buat batasan (min dan max) pada tanggal yang akan dipilih?



Exercise *for* Students

Latihan untuk *Students*

Create directory : C:\xampp\htdocs\lecture7\
Create file : latihan_.php

- ☐ Buat contoh dari metode **load()**, **get()**, **post()** dan **ajax()**, yang sudah di pelajari sebelumnya.
- ☐ Create an example of the **load()**, **get()**, **post()** and **ajax()** methods, we learned earlier.

Reference (#1)

Return JSON response from AJAX using jQuery and PHP

<https://makitweb.com/return-json-response-ajax-using-jquery-php/>

PHP - AJAX and PHP

https://www.w3schools.com/php/php_ajax_php.asp

PHP Ajax Tutorial with Example

<https://www.guru99.com/php-ajax.html>

Building a Simple Application with AJAX and PHP

https://subscription.packtpub.com/book/web_development/9781904811824/1/cho1vl1sec09/building-a-simple-application-with-ajax-and-php

Ajax (programming)

[https://en.wikipedia.org/wiki/Ajax_\(programming\)](https://en.wikipedia.org/wiki/Ajax_(programming))

PHP AJAX : Contoh AJAX dan XML

<https://dosenit.com/php/php-ajax-contoh-ajax-dan-xml>

Input Data Ke Database Tanpa Reload Dengan Ajax JQuery

<https://www.malasngoding.com/input-data-ke-database-tanpa-reload-dengan-ajax-jquery/>

jQuery Ajax POST dengan PHP

<https://qastack.id/programming/5004233/jquery-ajax-post-example-with-php>

Reference (#2)

Contoh real-time pencarian PHP AJAX

<http://www.w3big.com/id/php/php-ajax-livesearch.html>

Pengertian Proxy Beserta Fungsi

<https://www.nesabamedia.com/pengertian-proxy-dan-fungsi-proxy/>

Belajar Javascript: Apa itu AJAX?

<https://www.petanikode.com/javascript-ajax/>

Apa Itu AJAX dan Bagaimana Cara Kerjanya?

<https://www.hostinger.co.id/tutorial/apa-itu-ajax>

Belajar AJAX Dengan jQuery & PHP

<https://kelasprogrammer.com/belajar-ajax-dengan-jquery-php-untuk-pemula/>

jQuery ajax() Method

https://www.w3schools.com/jquery/ajax_ajax.asp

Ajax POST request with jQuery and PHP

<https://www.cluemediator.com/ajax-post-request-with-jquery-and-php>

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

